Testing the Fluorescence Advantage: Does Solar-Induced Fluorescence Signal Agricultural Drought Earlier than NDVI in Marathwada, India?

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Highlights

- Against normal years, GOSIF SIF and NDVI flagged drought within one 8-day composite.
- Neither index was consistently earlier, in 2015–2023 or in 2001–2014.
- SIF's "seasonal lead" over NDVI depends on how each curve is normalised.
- MODIS NIRv leads NDVI as much as GOSIF does; TROPOMI SIF led NIRv slightly in 2018–2020.
- India's rainfall Trigger-1, even applied per pixel, rarely fired in the drought years.

Abstract

Drought declarations in India, and the crop-insurance payouts (PMFBY) tied to them, rest on rainfall deficits and on vegetation indices such as NDVI. Solar-Induced Fluorescence (SIF), being tied to photosynthesis, is often expected to show drought stress earlier. I test that expectation for the eight districts of Marathwada, Maharashtra, over nine Kharif seasons (2015–2023), comparing cropland-weighted GOSIF SIF with MODIS NDVI dated to each pixel's observation day. The main question is whether SIF departs from its normal-year range earlier than NDVI in drought. Standardised against normal years, the two alarms almost always fell within one 8-day composite of each other: at the regional scale both crossed $z < -3$ on the same date in 2015 and NDVI was eight days earlier in 2018; across the 15 drought district-years (each district against its own rainfall baseline), SIF and NDVI flagged 12 and 13, and where both did, SIF was earlier in 7, they coincided in 2 and NDVI was earlier in 3, in all but one case by a single composite. At looser thresholds NDVI was earlier more often. Because NDVI is sampled every 16 days, the sign of a one-composite difference depends on how NDVI is placed in time: using only the NDVI observed by each date, SIF was first in both regional drought years and in 8 of the 12 district-years. A secondary question is whether SIF's normal seasonal decline starts before NDVI's. With each year scaled from its seasonal minimum to its peak, SIF crossed decline thresholds first in all nine years (8.0–34.2 days) and all 72 district-years. But NDVI in this region does not return to its pre-season level after the monsoon, so this scaling places NDVI's decline later by construction; scaled from peak to post-peak minimum, the lead falls to −5.5 to +13.4 days, SIF is first in six years, and the district differences disappear (Friedman $p = 0.97$). MODIS NIRv and EVI also declined before NDVI, by as much as GOSIF SIF did, and GOSIF did not lead NIRv (first in none of the nine years when both were scaled over their own decline); because GOSIF is predicted from MODIS reflectance, this is partly expected. TROPOMI SIF (2018–2020), which is not, declined before NDVI under both scalings and also before NIRv in all three years, by 1–13 days. Only 2015 and 2018 met the rainfall drought criterion, and the lead was not larger in them. Applied unchanged to 2001–2014 as an out-of-sample test with five more drought years, the methods gave the same picture: at district level the two alarms mostly coincided or differed by one composite in either direction (at $z < -1.5$, SIF earlier in 5 of 22 drought

district-years flagged by both, the same in 11, NDVI earlier in 6); regionally SIF flagged two of the five drought years at $z < -3$ and NDVI one, and the seasonal lead again matched NIRv's. An approximation of the rainfall-based Trigger-1, with dry spells counted inside the monsoon, never fired for the region and fired in 2 of 15 drought district-years, also when evaluated pixel by pixel. In this dataset SIF is not a resolvably earlier drought alarm than NDVI at the available sampling, GOSIF's seasonal lead is largely a property of how NDVI's decline is measured, and the small lead of direct TROPOMI SIF over NIRv needs more years to test.

1. Introduction

1.1 Motivation

NDVI is a reflectance-based measure: it tracks how much red and near-infrared light a canopy reflects, and it changes once leaf area, structure or chlorophyll content change. SIF works on a different physical principle. When a leaf absorbs sunlight, a small part of that energy is re-emitted as fluorescence, and the share depends on how the leaf is using the light. Photosynthetic efficiency can drop under stress before greenness does, so SIF should, in principle, show stress earlier than NDVI. Whether that physical head start appears as a measurable timing difference in the satellite products available for operational use is the empirical question this study tests.

India's drought-declaration process, and the PMFBY crop-insurance payouts tied to it, lean heavily on rainfall deficit and NDVI (Ministry of Agriculture & Farmers Welfare, 2016a, 2016b; Down To Earth, 2025). Two different things are often run together when SIF is proposed as an earlier indicator: SIF starting its normal seasonal decline before NDVI does, and SIF departing from its normal-year behaviour earlier than NDVI does in a drought year. Only the second would make SIF a better drought alarm. This study tests both, separately, and treats the second as the main question.

1.2 Previous Work and Research Gap

Satellite SIF responds to drought: GOME-2 and OCO-2 SIF fell during the 2011–2014 drought in Texas and California, in step with gross primary production (Sun et al., 2015), and in India's Indo-Gangetic Plains GOME-2 SIF showed the response of winter wheat to heat stress earlier than reflectance-based indices (Song et al., 2018). Whether it responds *before* greenness is less settled. Several studies find that much of the SIF signal at canopy scale follows absorbed light and canopy structure rather than fluorescence yield (Yoshida et al., 2015; Dechant et al., 2020), so SIF and a structural index such as NIRv (Badgley et al., 2017) can move together. Reviews of SIF for vegetation stress (Mohammed et al., 2019; Porcar-Castell et al., 2014) list early stress detection as a promise that depends on separating the yield signal from these structural effects.

Most studies of SIF lead times compare seasonal curves, or compare SIF and vegetation-index anomalies over large regions with coarse SIF retrievals. Few test, at the district scale used for drought declarations, whether a SIF *anomaly* precedes an NDVI anomaly in independently identified drought years, and few check how sensitive a "seasonal lead" is to the way each curve is normalised — a known issue in phenology extraction (Zhang et al., 2003; Jönsson & Eklundh, 2004), because NDVI and other reflectance indices saturate over dense canopies and differ in their post-season baseline (Huete et al., 2002). This study addresses both, for the district-level setting in which Indian drought declarations are made.

1.3 Physical Basis of Solar-Induced Fluorescence

A leaf absorbing photosynthetically active radiation splits that energy across three competing pathways: photochemistry (Φ_P), non-photochemical quenching or heat dissipation (Φ_{NPQ}), and chlorophyll fluorescence (Φ_F):

$$\Phi_P + \Phi_{NPQ} + \Phi_F = 1$$

Under well-watered conditions most absorbed energy goes to photochemistry and a small, fairly stable share is emitted as fluorescence. Under drought or heat stress, photochemistry is down-regulated and non-photochemical quenching increases; the fluorescence yield changes as this happens, before any change in canopy structure (Porcar-Castell et al., 2014; Figure 1).

The signal measured by a satellite is

$$SIF = APAR \times \Phi_F \times f_{esc}$$

where APAR is absorbed photosynthetically active radiation, Φ_F the fluorescence quantum yield and f_{esc} the fraction of emitted photons that escape the canopy. Because APAR and f_{esc} depend on canopy structure, much of the variation in canopy SIF can come from structure rather than from Φ_F (Dechant et al., 2020). Satellite instruments such as GOME-2, OCO-2 and TROPOMI retrieve SIF in Fraunhofer lines, where the small fluorescence contribution can be separated from reflected sunlight. The GOSIF product used here as the main SIF source is not a direct retrieval but a statistical reconstruction from OCO-2 soundings, MODIS reflectance and meteorological reanalysis (Li & Xiao, 2019). TROPOMI SIF, used as an independent check, is a direct retrieval.

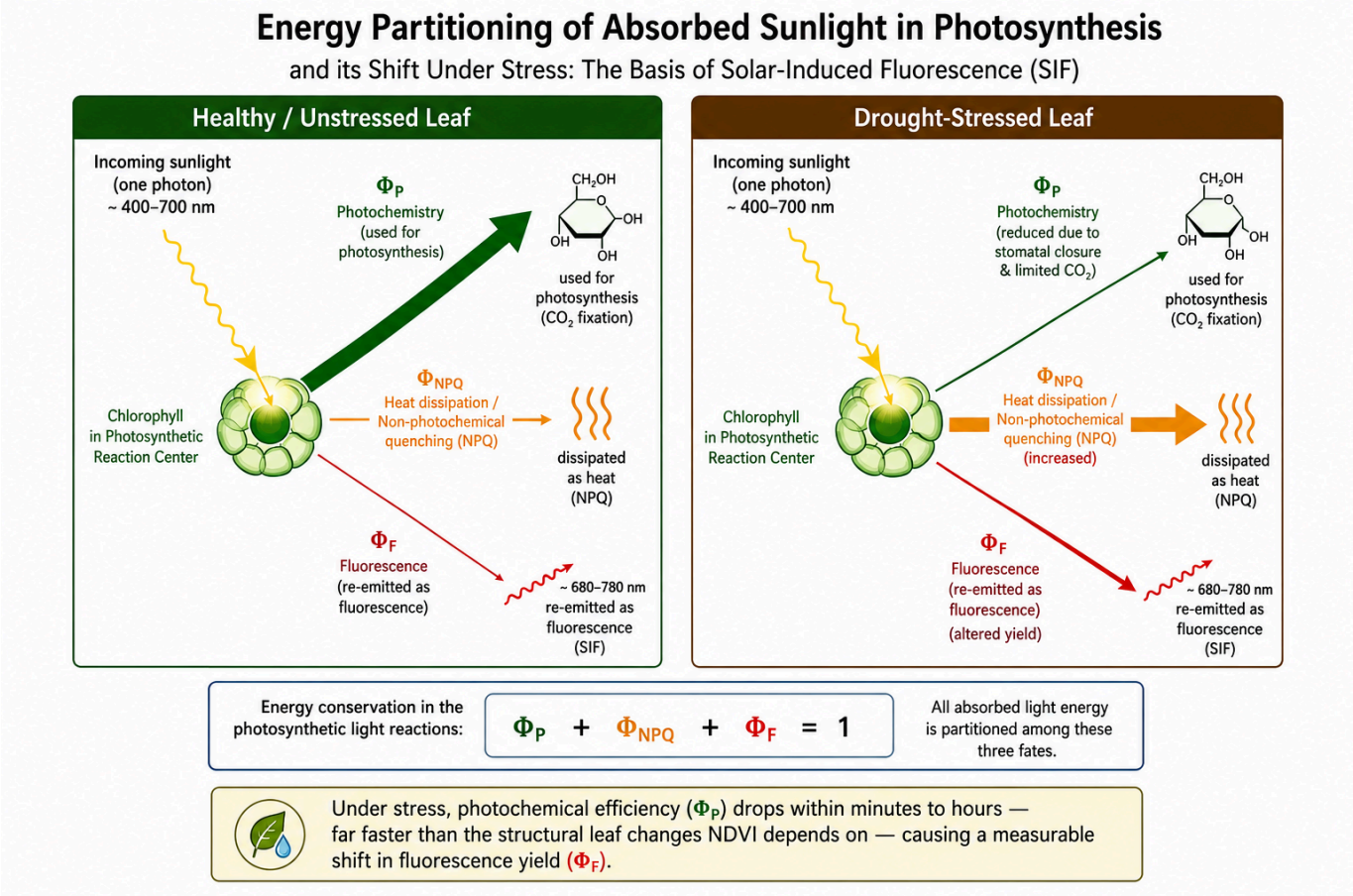


Figure 1. Schematic representation of leaf-level light energy partitioning among photochemistry, non-photochemical quenching (heat dissipation), and chlorophyll fluorescence, under well-watered versus drought-stressed conditions. *Illustration generated with an AI image tool, used only as a visual aid; it is not data or a result of this study.*

1.4 Physical Basis of NDVI

NDVI comes from surface reflectance in the red and near-infrared bands:

NDVI = (NIR − RED) / (NIR + RED)

Chlorophyll absorbs red light strongly, so red reflectance from healthy leaves is low, while the internal structure of the leaf — the air–cell-wall interfaces in the spongy mesophyll — scatters near-infrared light strongly (Figure 2). A dense, healthy canopy therefore has a large red/near-infrared contrast and a high NDVI; a thinning or senescing canopy narrows the contrast. NDVI saturates over dense canopies (Huete et al., 2002), so it can stay near its peak while leaf area is already falling. Cloud and cloud-shadow contamination is handled here by screening every pixel with MOD13Q1's `SummaryQA` flag (Section 3.1).

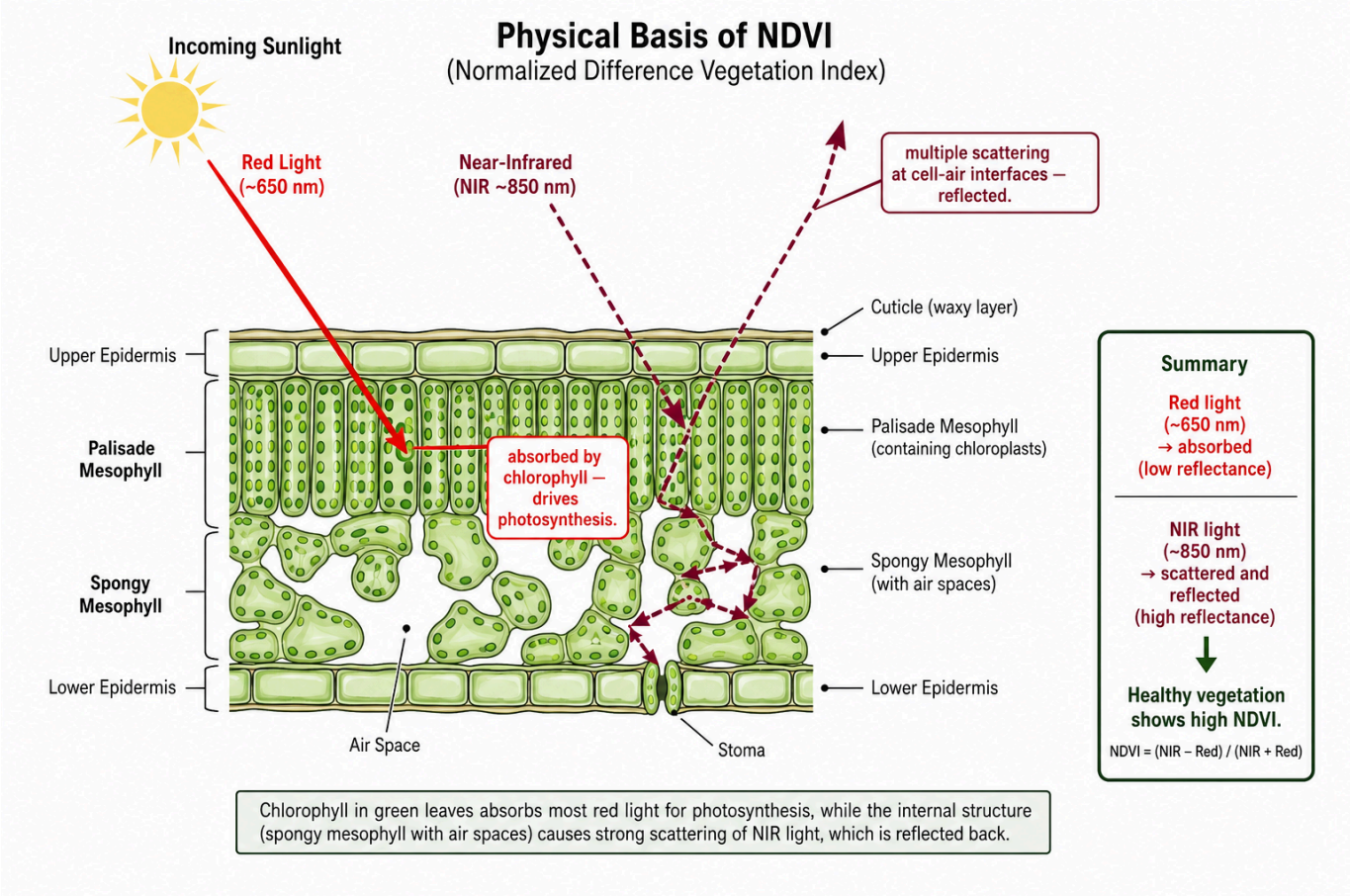


Figure 2. Leaf cross-section illustrating red-light absorption by chlorophyll in the palisade mesophyll versus near-infrared scattering at spongy mesophyll air-cell interfaces, the physical basis of the reflectance contrast measured by NDVI. *Illustration generated with an AI image tool, used only as a visual aid; it is not data or a result of this study.*

1.5 Research Questions and Hypotheses

Table 1. Research questions, hypotheses and where they are tested.

Research question	Hypothesis	Section
RQ1 (main): In drought years, does SIF depart from its normal-year range before NDVI does?	H4: SIF's drought anomaly precedes NDVI's.	3.9, 3.14, 4.2, 4.11
RQ1b: Does SIF's normal post-peak decline begin before NDVI's?	H1: SIF's seasonal decline begins before NDVI's.	3.4, 4.3–4.6, 4.11

Research question	Hypothesis	Section
RQ2: Does the SIF–NDVI lag differ between districts and with drought?	H2: the lag differs between districts. H3 (exploratory): drought widens the lag.	3.8, 4.7, 4.11
RQ3: How do satellite drought signals time against the rainfall Trigger-1 and the 2015 and 2018 declarations?	— (descriptive)	3.10, 4.9

H3 is exploratory because only two of the nine years (2015, 2018) meet the drought criterion (Section 3.5); the 2001–2014 out-of-sample test adds five more (Section 4.11).

2. Study Area

The study covers all eight districts of Marathwada, Maharashtra — Chhatrapati Sambhajinagar (Aurangabad), Jalna, Parbhani, Hingoli, Nanded, Beed (called "Bid" in the FAO GAUL dataset used for boundary definition (FAO, 2015); see Section 3.2), Latur, and Dharashiv (Osmanabad) — chosen for its long history of agricultural drought. Agriculture is mostly rain-fed Kharif cropping (soybean, cotton, pulses, sorghum) during the June–September monsoon.

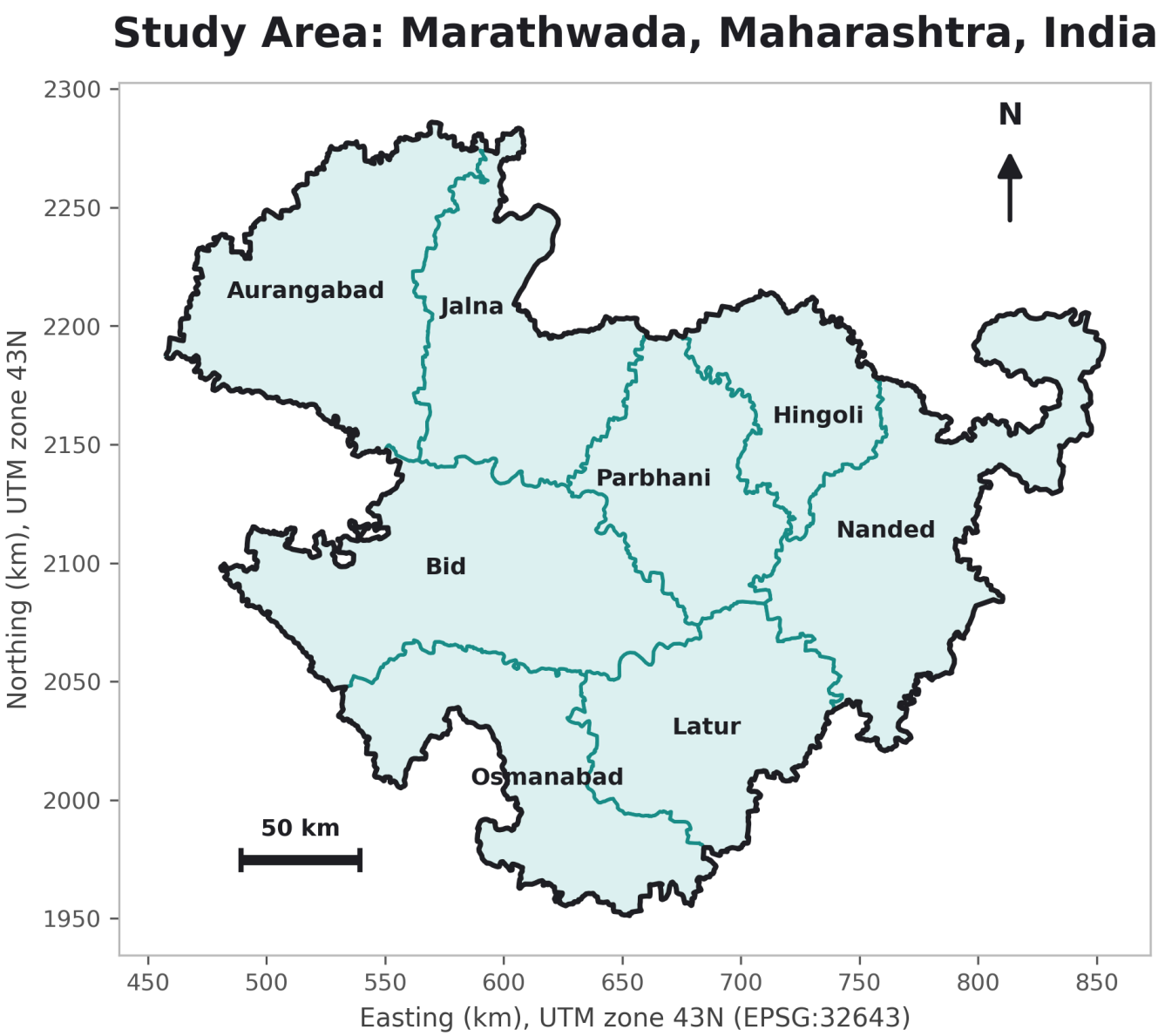


Figure 3. Study area: the eight districts of Marathwada, Maharashtra, India, with district boundaries and names.

3. Data and Methodology

3.1 Data Sources

All analyses cover the 1 June – 27 December window (day-of-year 153–361) of nine years, 2015–2023. The same window in 2001–2014 is used for the out-of-sample test of Section 3.14.

- **SIF (main):** GOSIF v2, a global 0.05°, 8-day solar-induced fluorescence product (Li & Xiao, 2019), distributed by the Global Ecology Group, University of New Hampshire.
- **SIF (independent check):** TROPOSIF v2.1 Level-2B daily soundings from Sentinel-5P TROPOMI (Guanter et al., 2021), for 2018–2020.
- **NDVI:** MODIS MOD13Q1 (Didan, 2021), 16-day, 250 m, via Google Earth Engine, including its per-pixel `DayOfYear` band (the day each pixel's value was observed).
- **Land cover:** MODIS MCD12Q1 (Friedl & Sulla-Menashe, 2022), IGBP classes 12 (cropland) and 14 (cropland/natural vegetation mosaic).
- **Precipitation:** CHIRPS daily rainfall (Funk et al., 2015), via Google Earth Engine: seasonal totals for 1981–2023 at region and district level, and daily series for the region (2001–2023) and districts (2015–2023).

Every rainfall anomaly in this study comes from this one series.

- **Crop yields:** district-season-crop Area/Production/Yield statistics (Ministry of Agriculture & Farmers Welfare, 2024), Kharif season, 1997-98 to 2022-23.
- **Official declarations:** the 2015 Maharashtra drought declaration (reported 16 October 2015; India.com/ANI, 2015), the 2018 first-stage declaration of a drought-like situation (23 October 2018; Deccan Herald, 2018), and the 2018 final Kharif drought declaration (31 October 2018; Zee News, 2018; Economic and Political Weekly, 2018).

Three properties of this dataset combination shape how the results should be read. First, GOSIF is trained on OCO-2 SIF with MODIS reflectance (including EVI) and MERRA-2 reanalysis as predictors (Li & Xiao, 2019). Because MODIS reflectance also underlies MOD13Q1 NDVI, GOSIF and NDVI are not independent measurements; this is why the TROPOMI check (Section 3.11) was added. Second, GOSIF's 0.05° grid (about 5 km) is far coarser than MOD13Q1's 250 m, so every comparison is a regional- or district-scale signal, not a field-scale one. Third, class 14 is a cropland/natural-vegetation mosaic, so "cropland" here means a mixed agricultural landscape; classes 12 and 14 together cover about 96–97% of the study area's cells.

3.2 Boundary Definition and Verification

The study boundary comes from the FAO GAUL 2015 level-2 dataset, filtered to the eight Marathwada districts and merged into one polygon. Both the SIF clipping and the NDVI extraction use this polygon. Overlaying it on a clipped SIF raster confirmed that data survive only inside the district outlines (Figure 4).

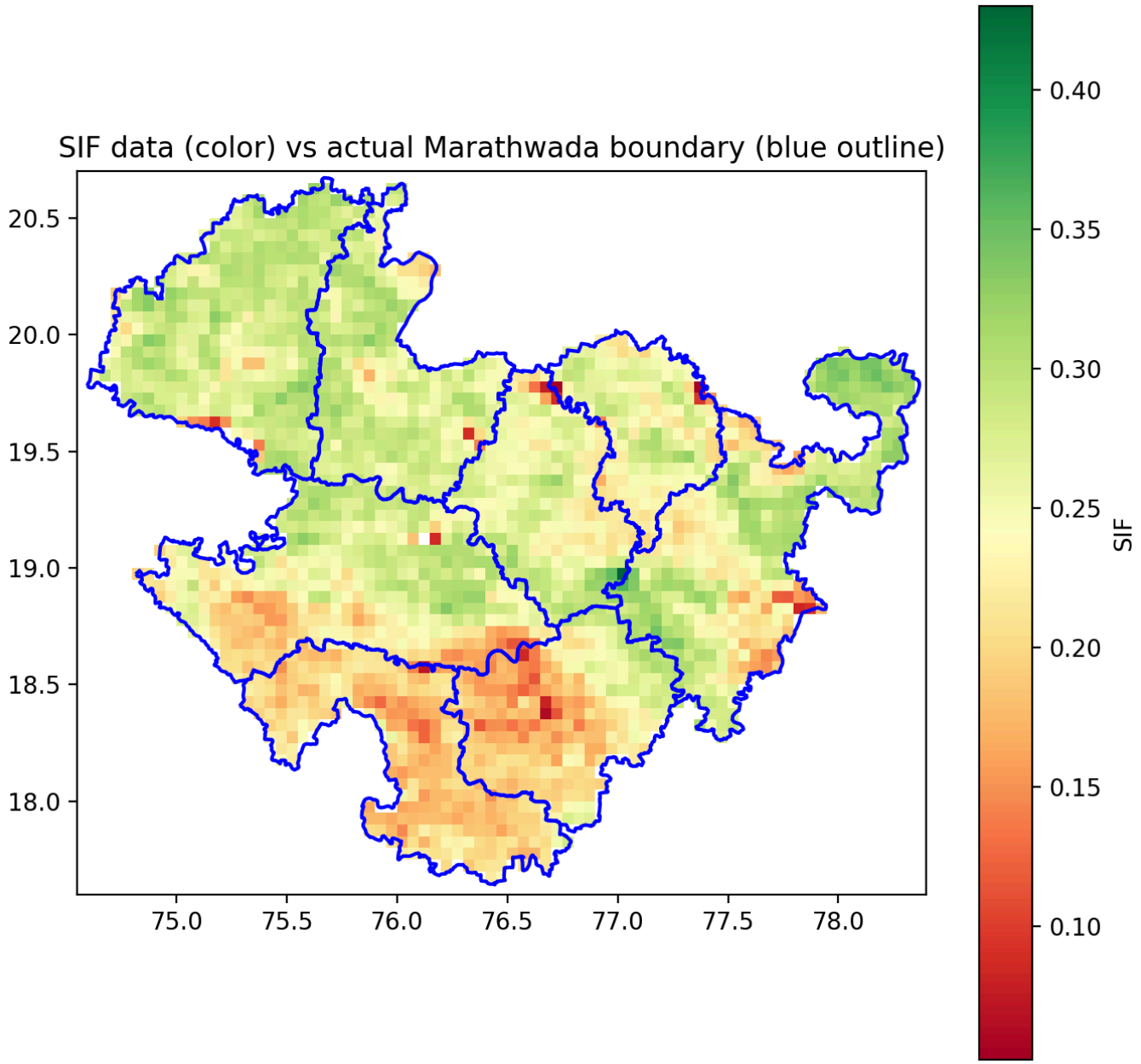


Figure 4. Verification of boundary masking: clipped SIF data (color) overlaid with the Marathwada district boundary (blue outline).

3.3 Processing Pipeline

SIF. GOSIF GeoTIFFs were clipped to the study boundary, scaled by GOSIF's 0.0001 factor, and masked for fill values (32767 water, 32766 snow/ice or barren). Each 0.05° cell was weighted by its cropland fraction — the share of MCD12Q1 pixels in classes 12 or 14 within that cell, computed in Earth Engine on the GOSIF grid. Each 8-day composite is dated at its centre (start day + 3.5).

NDVI. MOD13Q1 pixels were kept only where `SummaryQA` marks good or marginal quality and where MCD12Q1 classifies the pixel as cropland. Because a 16-day MOD13Q1 value can come from any day in its window, each composite was dated by the mean of its pixels' `DayOfYear` values rather than by its start date. Each year's NDVI series was normalised on its own observations and then linearly interpolated onto the SIF composite dates. During the monsoon many NDVI composites rest on few clear pixels (23 of 117 regional composites have under 25% of cropland pixels passing the quality screen); excluding such composites does not

change the direction of any year's lag, although with a stricter 50% cut-off the 2022 lag turns negative (−6.2 days; Supplementary Figure S1).

3.4 Seasonal Lag Calculation

Each year's SIF and NDVI series was rescaled to 0–1 and the day-of-year on which each series first fell below a set of decline thresholds after its peak (0.9, 0.8, 0.7, 0.6, 0.5) was found by linear interpolation. The lag at each threshold is the NDVI crossing date minus the SIF crossing date (positive = SIF first), and a year's lag is the mean over the thresholds it reaches. Two rescalings were used:

- **Season range** (the original method): 0 at the year's seasonal minimum, 1 at its peak. The minimum is usually the pre-monsoon value in June.
- **Decline range**: 0 at the minimum *after* the peak, 1 at the peak, so each series is measured over its own decline.

The two give the same result if both series return to their pre-season level at the end of the season. If one does not, the season-range scaling makes that series cross every threshold later, because its post-season floor sits partway up the scale. An asymmetric double-logistic fit to each series, with the end-of-season inflection date as a third estimate, was also computed but is unstable with 8- and 16-day data and is reported only as a sensitivity check (Supplementary Figure S2).

To test whether any lead that remains is a property of NDVI rather than of fluorescence, I repeated the same threshold-crossing comparison with two other MOD13Q1 reflectance indices from the same composites and the same quality and cropland masking: EVI, and NIRv (NDVI × near-infrared reflectance), a proxy for canopy structure and absorbed light (Badgley et al., 2017). SIF was compared with each index, and EVI and NIRv were compared with NDVI on their shared observation dates.

3.5 Rainfall Anomaly and Drought Classification

Seasonal CHIRPS totals for each year (1 June – 26 December) were compared against a 2001–2020 mean and standard deviation, giving a percentage anomaly and z-score. A year is a drought year when its z-score is below −0.5. June–September totals give the same two drought years. District-years were classified against each district's own 2001–2020 baseline (main) and, as a check, against the regional baseline. Because the 2001–2020 baseline contains six study years, the regional classification was repeated with three other baselines (2001–2014, 1981–2014, 1991–2020).

3.6 Cross-Correlation Lag

As a scale-free estimate, I used time-lagged cross-correlation over the post-peak decline window, starting at SIF's peak. Both series were interpolated to a daily grid, de-meaned, and cross-correlated at shifts from −N/4 to +N/4 days, N being the length of the decline window; the shift maximising the correlation between SIF(t) and NDVI(t + lag) is that year's estimate (positive = SIF first). Because correlation does not depend on scale, this estimate is not affected by the choice of normalisation.

3.7 Uncertainty

For cross-correlation, each year's observations were case-resampled 2,000 times and re-run through the same search; the 2.5th–97.5th percentiles give a 95% interval. For threshold-crossing, each series was smoothed (Savitzky–Golay, window 5, order 2), the residuals were resampled in moving blocks of three composites (to keep their autocorrelation) onto the smooth curve 2,000 times, and the lag recomputed; replicates were

centred on the reported lag rather than on the smoothed curve's lag. A leave-one-observation-out jackknife gives a second view. The per-year draws were also combined into an interval for the drought-minus-normal difference that reflects measurement uncertainty only, not year-to-year variability.

3.8 District-Level Lag (H2, H3)

The same lags were computed for every district-year (8 × 9 = 72). Whether the lag differs between districts was tested with a two-way ANOVA (district and year effects) and a Friedman test across the nine years; drought and normal district-years were compared with Welch's t-test.

3.9 Drought-Specific Anomaly Signal (H4)

Seasonal lags say when each index starts its normal decline, not when it departs from normal. For each composite date, each year's SIF (and NDVI) value was standardised against the normal years at that date, excluding the year itself: z = (x − mean of normal years) / s.d. of normal years. A year's alarm date for a threshold (−0.5, −1, −1.5, −2, −3) is the first composite from 1 July at which z stays below the threshold for two consecutive composites. At the regional scale, $z < -3$ is the only threshold at which neither index raises an alarm in any normal year, so it is the main alarm threshold. The same procedure was applied to the 72 district-years: 15 drought and 57 normal against each district's own rainfall baseline (main), and 17 and 55 against the regional baseline (sensitivity). The separation of drought from normal district-years was also summarised date by date with the area under the ROC curve (AUC). District-years share each year's weather, so the AUCs are descriptive.

NDVI is placed on the SIF dates by linear interpolation between its observation days, so the NDVI value on a date already contains part of the next observation, up to about 16 days later. To see how much the comparison depends on this, the alarm test was repeated with NDVI on each SIF date taken as the last value observed by the end of that SIF composite. Because both the SIF composites and the interpolated NDVI are spaced 8 days apart, alarm differences of 8 days or less are reported as within one composite.

3.10 Rainfall Trigger and Declaration Dates (RQ3)

India's Manual for Drought Management (Ministry of Agriculture & Farmers Welfare, 2016a) starts drought assessment with a rainfall-based Trigger-1 built on rainfall deviation and dry spells. I approximated it as the first day after 1 July on which cumulative rainfall since 1 June is more than 25% below the 2001–2020 normal for that date and a dry spell (consecutive days below 2.5 mm) of at least 21 days has occurred, with dry spells counted only between 1 July and 30 September. Variants (deficit > 20% or > 30%, dry day < 1 mm, deficit alone, dry spell alone, and dry spells counted to December) were also run. The state applies the trigger at taluka level, and averaging rain over a district rarely leaves a 21-day spell, so I also evaluated the same rule in every 0.05° CHIRPS pixel and recorded, for each district-year, the share of the district's pixels in which it fired and the first day on which that share reached 25% and 50%. Both are approximations of the taluka-level procedure. Satellite alarm dates were compared against this trigger date and against the three declaration dates.

3.11 Independent Check with TROPOMI SIF

TROPOSIF soundings over Marathwada (1 June – 27 December, 2018–2020) were kept where cloud fraction < 0.2, solar zenith angle < 70° and the product's land-cover flag is cropland (12 or 14), and averaged into 8-day composites aligned with GOSIF's. The main variable is SIF_Corr_743 (Guanter et al., 2021); three variants were also run (cloud fraction < 0.5, all land cover, uncorrected SIF_743). The seasonal lag against NDVI was computed under both rescalings, and in the same way against MODIS EVI and NIRv (Section 3.4).

3.12 Method Dependence

The regional lag was recomputed under 40 combinations: SIF cropland-weighted or unweighted; NDVI timed by observation date or by compositing window; raw or smoothed series; and lag metric mean-threshold, 0.9, 0.7, 0.5, or cross-correlation (all with season-range scaling). For each combination I recorded whether SIF leads in all nine years, whether drought years have a smaller mean lag than normal years, and whether 2018 has the smallest lag.

3.13 Spatial Autocorrelation and Crop-Yield Check

Moran's I was computed each year for district SIF and rainfall anomaly with Queen-contiguity weights (9,999 permutations). For the yield check, every district-crop pair with at least 10 of the 26 Kharif seasons was converted to yield z-scores against its own mean and standard deviation and averaged into a Yield Anomaly Index per district-year and per region-year (n = 8 region-years, as the yield series ends in 2022-23).

3.14 Out-of-Sample Test, 2001–2014

After the analysis of 2015–2023 was complete, I applied it without change to 2001–2014, the GOSIF years before the study period. The drought rule, the alarm rule and its thresholds, the cropland weighting (with each year's MCD12Q1 map), the NDVI quality screen and observation-day dating, and both lag scalings are those of Sections 3.3–3.9; nothing was re-tuned on the new years. Anomalies are standardised against the normal years of 2001–2014 only, so no vegetation data from 2015–2023 enters the test; drought years are still defined against the 2001–2020 rainfall baseline, as in the main analysis. Before extracting the new years, I re-extracted 2015 with the same code and it reproduced the main series exactly (largest difference about 10^{-16}). TROPOMI does not cover these years, and the district-level lag tests (H2) were not repeated.

4. Results

4.1 Rainfall and Drought Years

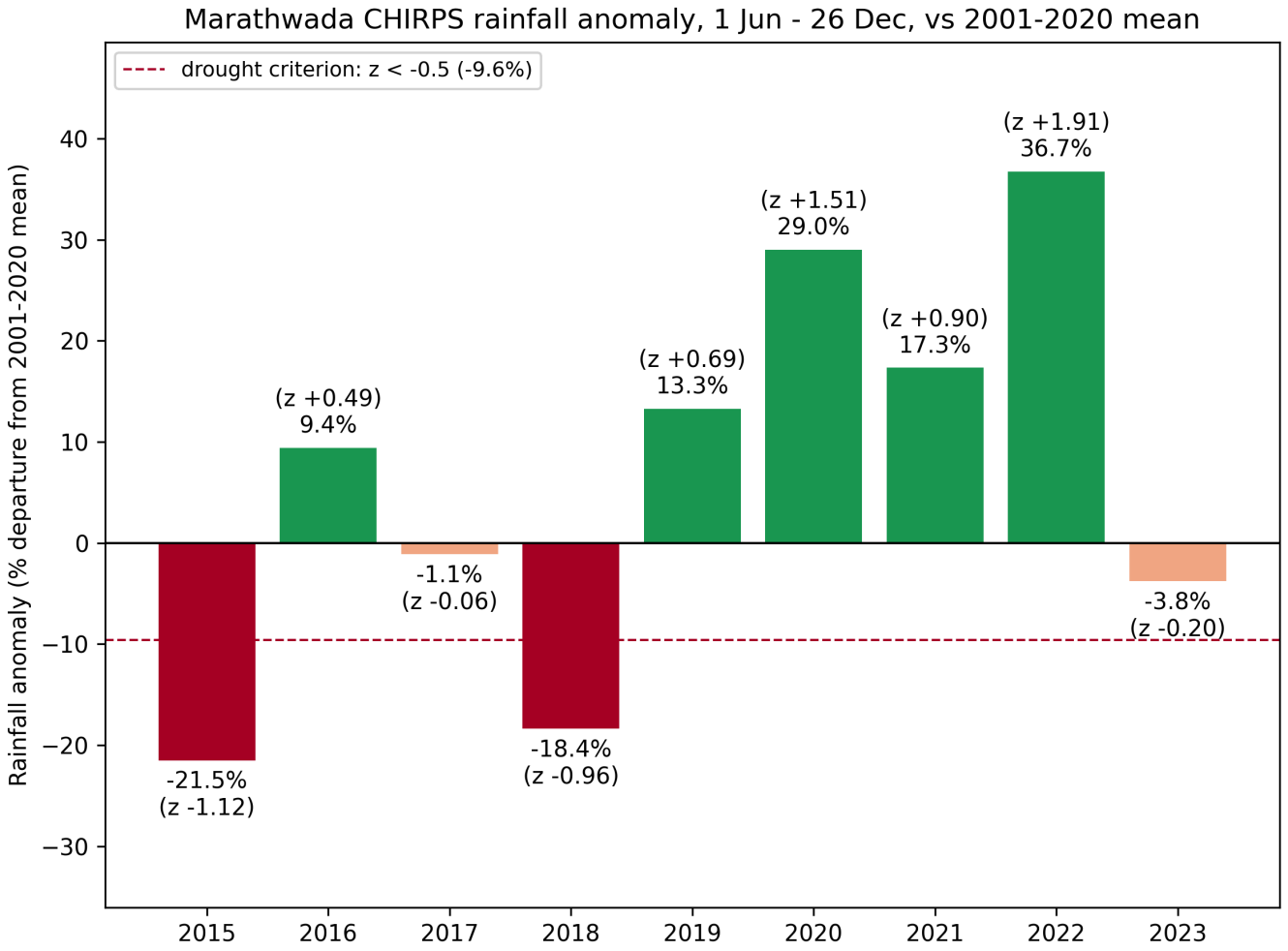


Figure 5. Regional June–December rainfall anomaly (% departure from the 2001–2020 mean), 2015–2023, with z-scores; dark red = drought years (z < −0.5).

Against a 2001–2020 mean of 826.8 mm (σ = 158.9 mm), the nine years came in at 649.0 mm in 2015 (−21.5%, z = −1.12); 904.7 mm in 2016 (+9.4%, z = 0.49); 817.4 mm in 2017 (−1.1%, z = −0.06); 674.9 mm in 2018 (−18.4%, z = −0.96); 936.3 mm in 2019 (+13.3%, z = 0.69); 1066.5 mm in 2020 (+29.0%, z = 1.51); 970.0 mm in 2021 (+17.3%, z = 0.90); 1130.3 mm in 2022 (+36.7%, z = 1.91, the wettest season since 1981); and 795.3 mm in 2023 (−3.8%, z = −0.20). Only 2015 and 2018 meet the drought criterion, under all four baselines; with the 2001–2020 baseline this holds for every cut-off from −0.3 to −0.75, while with the 1981–2014 baseline 2018 (z = −0.65) drops out at cut-offs of −0.7 and below. 2023 is a borderline case: its seasonal total is near normal, but August had 51 mm (z = −2.52), and Maharashtra declared drought in 40 talukas on 31 October 2023 (Morung Express/IANS, 2023). Under the same criterion, 2001, 2002, 2003, 2004 and 2014 are also drought years; Section 4.11 uses them as an out-of-sample test.

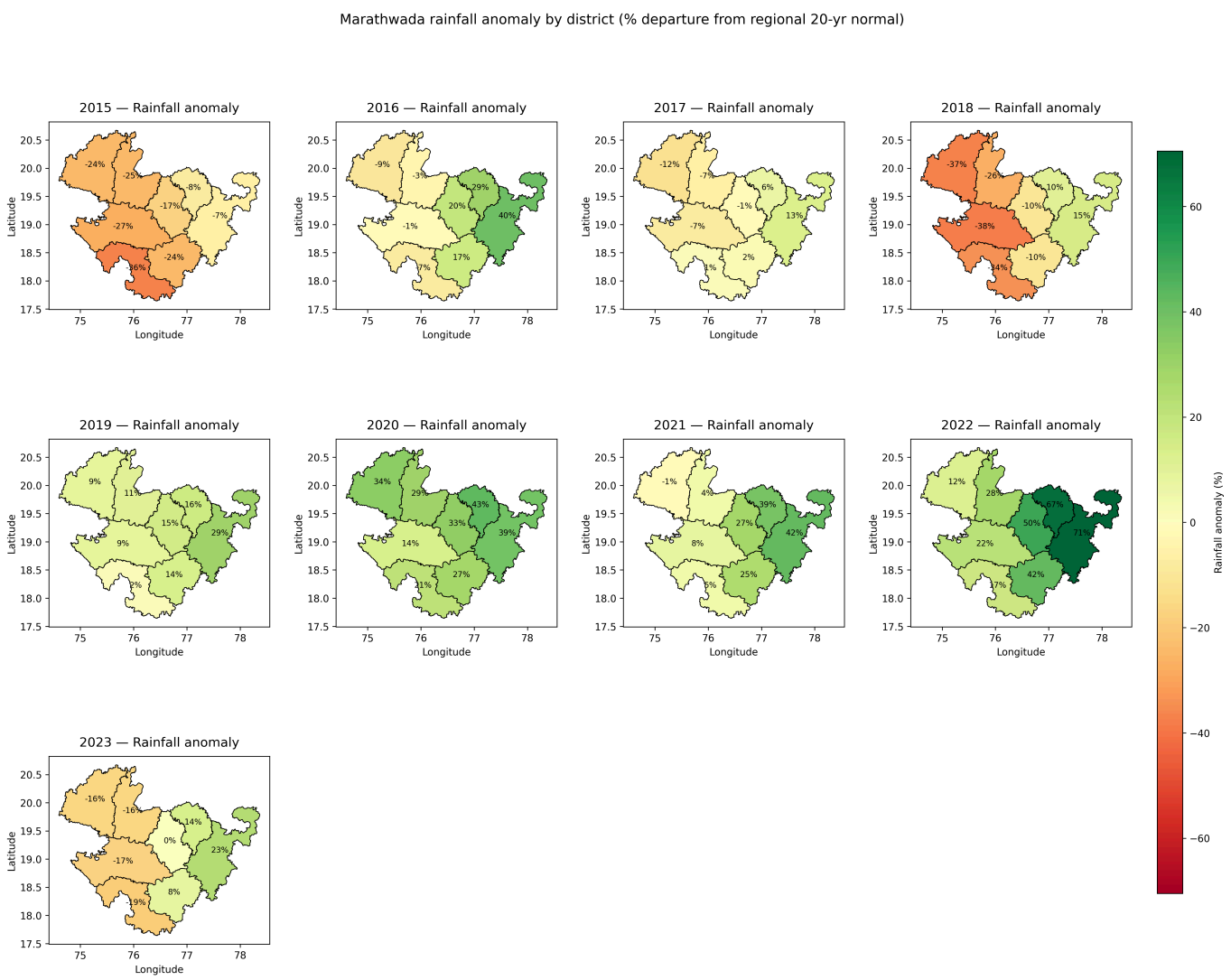


Figure 6. Rainfall anomaly by district (% departure from the regional 2001–2020 mean), 2015–2023.

The deficit was not spatially uniform. In 2018, Bid (−38.0%), Aurangabad (−37.3%) and Osmanabad (−33.9%) took the largest deficits while Hingoli (+10.2%) and Nanded (+14.6%) had surpluses; in 2015, Osmanabad (−36.4%) had the largest deficit and Hingoli (−8.0%) and Nanded (−7.1%) the mildest. Against each district's own baseline, 15 district-years are drought years: all eight districts in 2015, six in 2018 (all but Hingoli and Nanded) and Jalna in 2023.

4.2 Is There a Drought-Specific SIF Advantage? (H4)

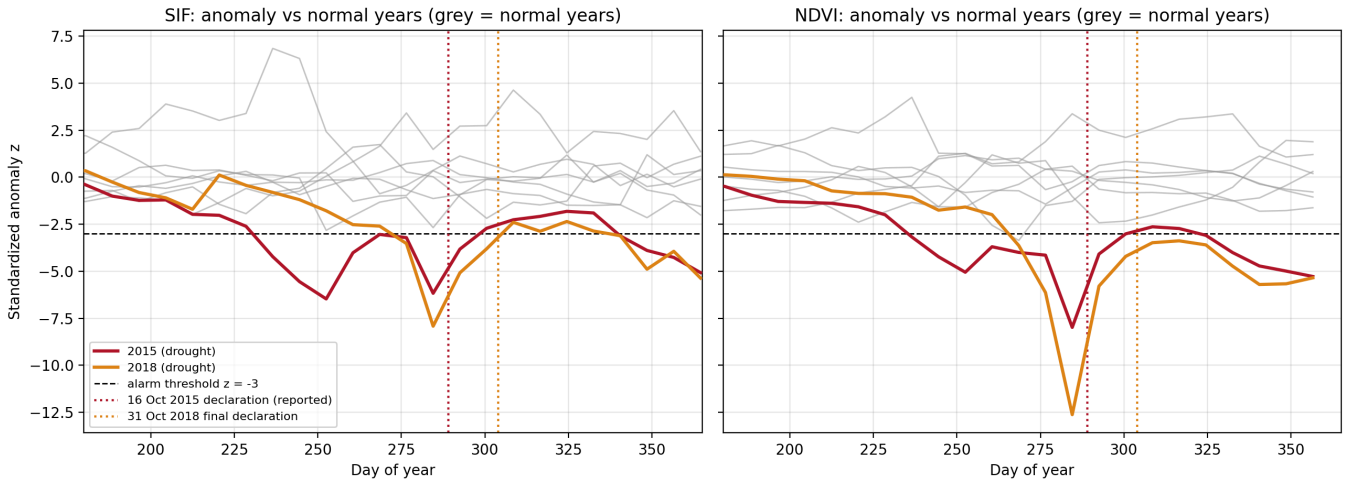


Figure 7. Standardised SIF and NDVI anomalies relative to normal years, by composite date; 2015 red, 2018 orange, normal years grey. The dashed line marks the alarm threshold z = −3; the dotted vertical lines mark the 2015 declaration and the 2018 final declaration.

At the regional scale, at z < −3, SIF and NDVI both raised the 2015 alarm on 24 August; in 2018 NDVI did so on 25 September and SIF on 3 October, one composite (eight days) later. At z < −1, SIF came first in both drought years (by 8 days in 2015 and 32 in 2018), but at that threshold each index also raised alarms in five of the seven normal years, so it does not separate drought from normal seasons. Both indices reached their deepest anomaly in 2018 on the same composite (11 October), NDVI more strongly (z = −12.6 vs −7.9).

Table 2. Drought alarms in district-years, own-district rainfall baseline (15 drought, 57 normal district-years).

Threshold	Drought flagged, SIF	Drought flagged, NDVI	False alarms, SIF	False alarms, NDVI	Both flagged	SIF earlier	Same date	NDVI earlier	Within one composite
z < −1.5	15	15	22	24	15	2	4	9	9
z < −2	14	15	11	12	14	4	4	6	9
z < −3	12	13	2	4	12	7	2	3	11

At the main threshold SIF was earlier in 7 of the 12 drought district-years flagged by both, the two coincided in 2 and NDVI was earlier in 3; in 11 of the 12 the difference was at most one composite. At looser thresholds NDVI was earlier more often (9 against 2 at z < −1.5, 6 against 4 at z < −2), again mostly by one composite (Table 2). SIF had slightly fewer false alarms. With the regional baseline instead (17 drought district-years), at z < −3 both flag 12, SIF is earlier in 5, the same in 4 and later in 3. The date-by-date AUC shows SIF separating drought from normal district-years from 23 July (AUC 0.80) and NDVI from 31 July (0.76). SIF's AUC is higher from 15 to 31 July and again on 24 August–1 September, NDVI's in late June–early July (when both are near 0.5) and on 8–16 August; from 9 September to 20 November the two differ by at most about 0.02, and by up to 0.06 in late November and early December.

These differences are at or below the sampling of the data, and their direction depends on how NDVI is placed in time (Section 3.9). Using only the NDVI observed by the end of each SIF composite, SIF was first at z < −3 in both regional drought years (by 8 days each) and in 8 of the 12 district-years flagged by both (same date in 3, NDVI earlier in 1).

H4 is therefore not supported in the sense the study set out to test: SIF did not give a consistently earlier drought alarm that these data can resolve, and neither did NDVI. The two alarms were usually within one composite of each other, the direction of that difference changes with the NDVI dating, and SIF's only consistent advantage is slightly fewer false alarms.

4.3 Seasonal SIF–NDVI Trajectories

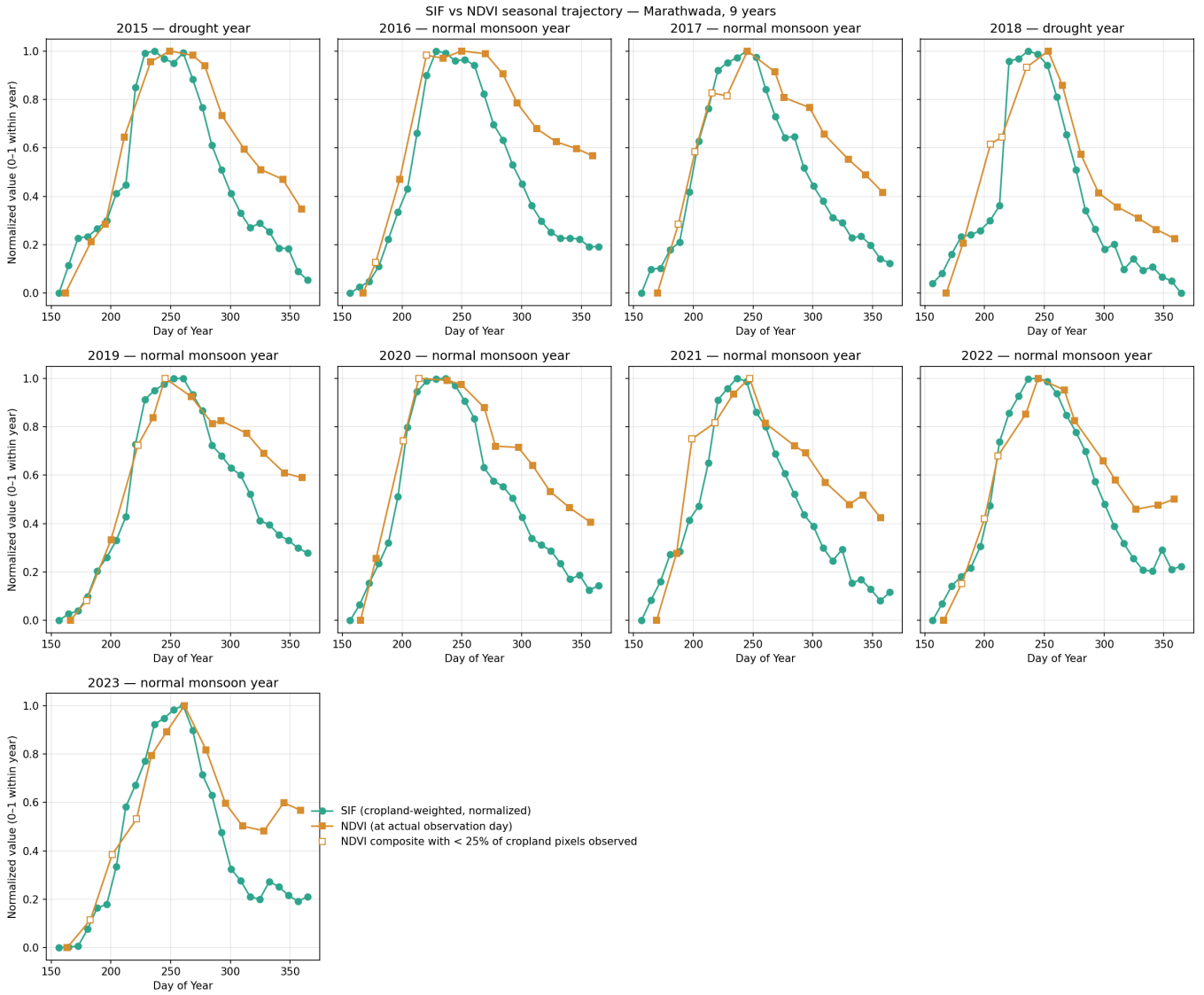


Figure 8. Normalised cropland-weighted SIF and NDVI (at observation day) seasonal trajectories, Marathwada, 2015–2023 (season-range scaling). Open squares mark NDVI composites in which fewer than 25% of cropland pixels passed the quality screen.

Under season-range scaling, SIF's post-peak decline appears to get under way ahead of NDVI's in all nine years, with NDVI holding near its peak for longer. Figure 8 also shows why the scaling matters: by late December SIF is back near its June level (0.00–0.28 on the season scale), but NDVI is not (0.23–0.59). Post-monsoon NDVI stays well above its pre-monsoon level, likely because of residual vegetation, Rabi sowing and weeds on fallow land.

4.4 Threshold-Crossing Lag and Its Dependence on Normalisation

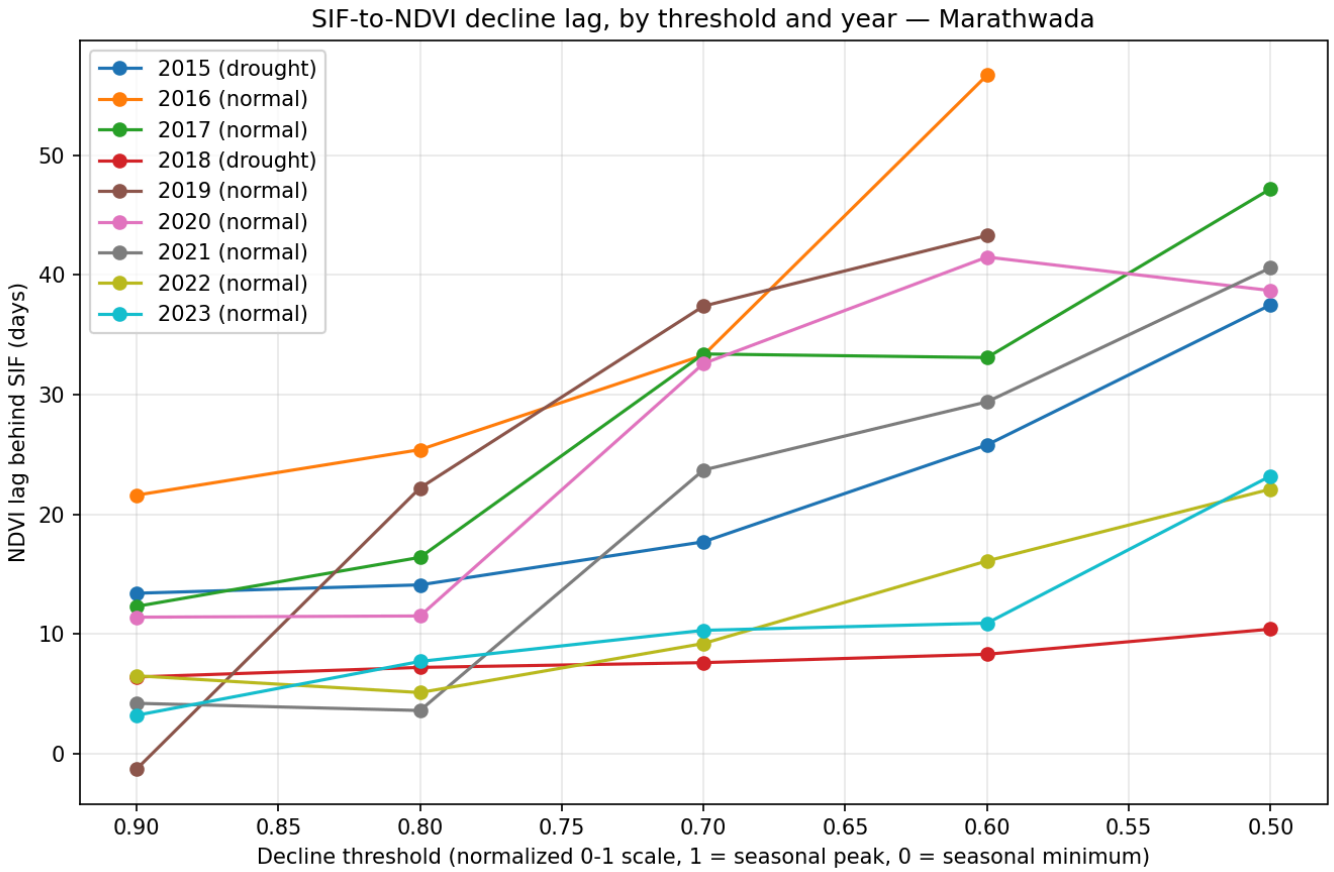


Figure 9. SIF-to-NDVI decline lag (days), by decline threshold and year (season-range scaling).

Under season-range scaling, the lag was 21.7 days (2015), 34.2 (2016), 28.5 (2017), 8.0 (2018), 25.4 (2019), 27.1 (2020), 20.3 (2021), 11.8 (2022) and 11.1 (2023), all positive, and the lag grows with depth of decline (Figure 9). The moving-block bootstrap puts every year above zero (95% intervals: 2015 [16.3, 30.4]; 2016 [12.5, 39.6]; 2017 [17.5, 32.5]; 2018 [2.8, 16.1]; 2019 [8.9, 33.0]; 2020 [18.6, 34.4]; 2021 [4.0, 25.2]; 2022 [5.3, 14.9]; 2023 [7.6, 16.6]); 8 of the 36 pairs of years have non-overlapping intervals.

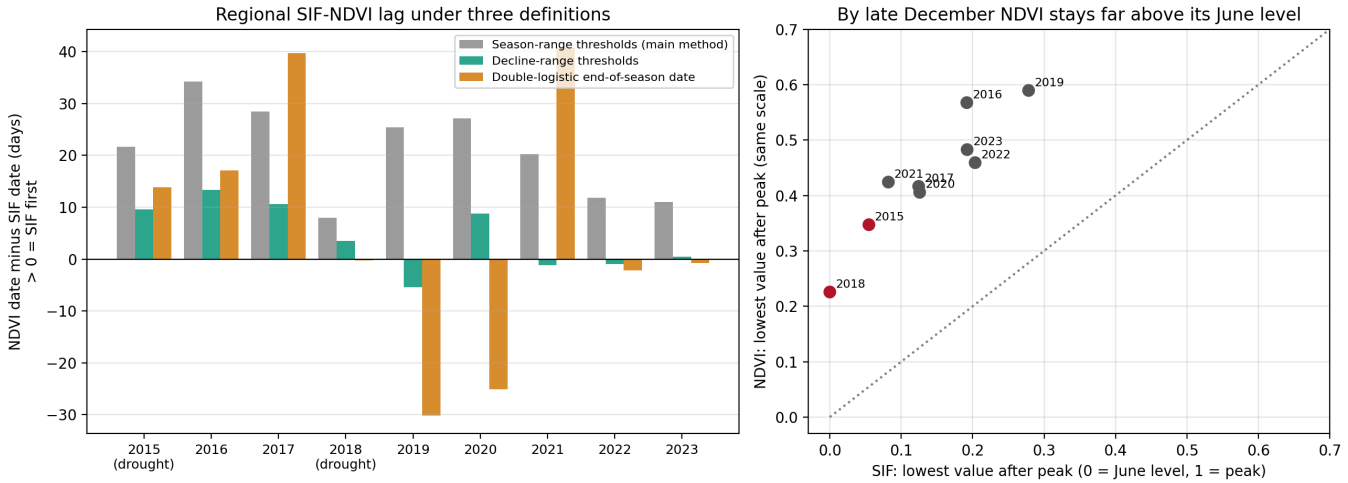


Figure 10. Regional SIF–NDVI lag by year under season-range scaling, decline-range scaling, and the double-logistic end-of-season date.

That lead rests on the scaling. When each series is scaled from its peak to its own post-peak minimum, the lag becomes 9.6 (2015), 13.4 (2016), 10.6 (2017), 3.5 (2018), −5.5 (2019), 8.8 (2020), −1.2 (2021), −1.0 (2022) and 0.4 (2023) days: SIF is first in six years by 0.4–13.4 days, and NDVI is first by 1–6 days in 2019, 2021 and 2022

(Figure 10). The double-logistic end-of-season date gives lags from −30 to +41 days with no consistent sign, and its fits are unstable (Supplementary Figure S2). How far each year's lead shrinks broadly follows how much higher NDVI's post-season floor sits than SIF's: across the nine years, that gap (on the season scale) correlates with the difference between the two lags at r = 0.67.

Table 3. Regional results by year and method.

Year	Drought?	Season-range lag (days)	Block-bootstrap 95% interval	Decline-range lag (days)	Cross-correlation lag (days)	Cross-correlation 95% interval
2015	Yes	+21.7	[16.3, 30.4]	+9.6	+10	[6, 13]
2016	No	+34.2	[12.5, 39.6]	+13.4	+12	[8, 15]
2017	No	+28.5	[17.5, 32.5]	+10.6	+14	[−20, 28]
2018	Yes	+8.0	[2.8, 16.1]	+3.5	+4	[0, 6]
2019	No	+25.4	[8.9, 33.0]	−5.5	0	[−20, 24]
2020	No	+27.1	[18.6, 34.4]	+8.8	+9	[−28, 28]
2021	No	+20.3	[4.0, 25.2]	−1.2	+20	[−4, 30]
2022	No	+11.8	[5.3, 14.9]	−1.0	−1	[−5, 3]
2023	No	+11.1	[7.6, 16.6]	+0.4	−3	[−10, 0]

4.5 Cross-Correlation and Its Uncertainty

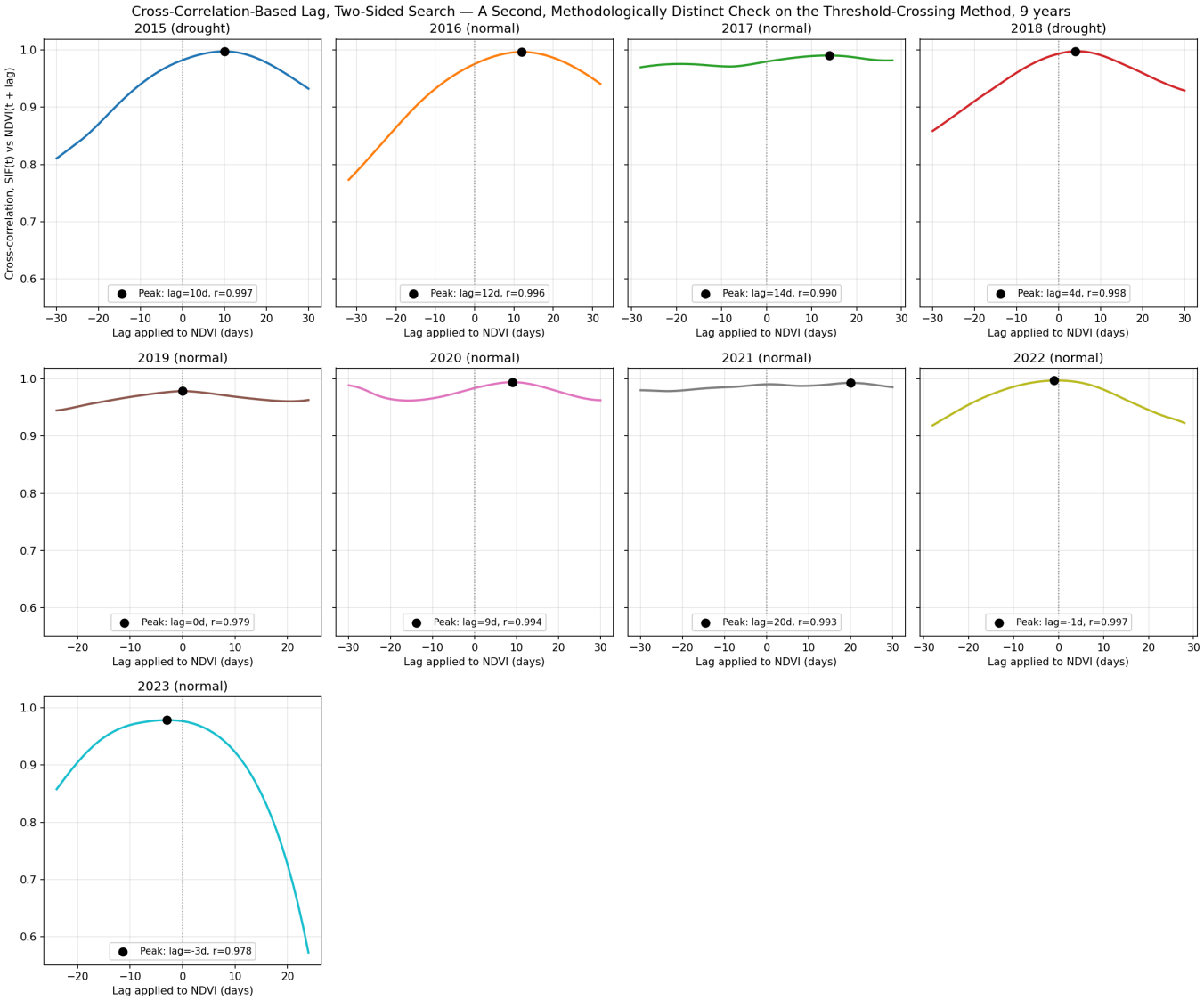


Figure 11. Cross-correlation between SIF and NDVI as a function of the lag applied to NDVI, by year. Black markers show the correlation-maximising lag; a negative lag means NDVI leads SIF.

Cross-correlation does not depend on scaling. The correlation-maximising shift was 10 days (2015), 12 (2016), 14 (2017), 4 (2018), 0 (2019), 9 (2020), 20 (2021), −1 (2022) and −3 (2023), with peak correlations of 0.979–0.998. The correlation curves are flat — the correlation at zero shift is already 0.975–0.993 — so the shift is poorly determined.

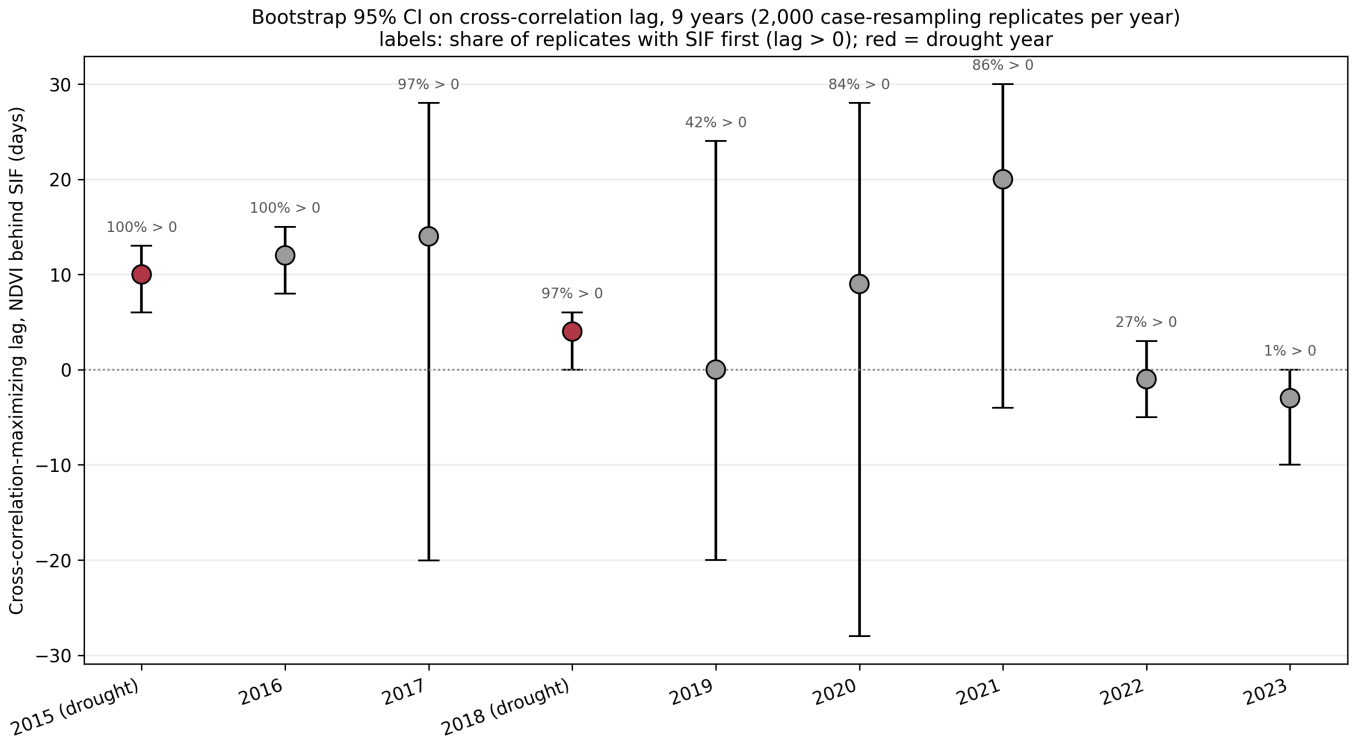


Figure 12. Cross-correlation lag point estimates with 95% bootstrap intervals (case resampling, 2,000 replicates per year), labelled with the percentage of replicates showing a SIF lead.

The 95% intervals exclude zero on the SIF-lead side only in 2015 and 2016; 2018 sits on zero, and 2023 leans the other way (Table 3). Cross-correlation and the decline-range lag broadly agree: both put 2015 and 2016 among the clearest SIF leads and find no lead in 2019, 2022 and 2023. The two scale-independent or scale-matched methods therefore point to a small, year-dependent SIF lead, not the consistent three-week lead of the season-range method.

4.6 Independent Check with TROPOMI SIF

Table 4. TROPOMI (TROPOSIF v2.1) versus GOSIF, 2018–2020. Lags are NDVI crossing minus SIF crossing (days).

Year	Composites	r (TROPOMI vs GOSIF)	Peak DOY TROPOMI	Peak DOY GOSIF	Season-range lag, TROPOMI	Season-range lag, GOSIF	Decline-range lag, TROPOMI	Decline-range lag, GOSIF
2018	26	0.91	244.5	236.5	12.8	8.0	10.4	3.5
2019	26	0.97	252.5	260.5	43.8	25.4	10.3	−5.5
2020	26	0.97	212.5	236.5	45.3	27.1	27.7	8.8

Of 630,091 soundings inside Marathwada on 540 days, 53% passed the cloud filter. TROPOMI reproduces GOSIF's seasonal trajectory closely (r = 0.91–0.97) and puts SIF's decline before NDVI's in all three years under both scalings, though by different amounts; the four TROPOMI variants agree (season-range lags 10.8–45.9 days, none negative). TROPOMI peaks earlier than GOSIF in 2020 and its lead is larger, so the size of the lead does not carry over between products. The 2018 deficit relative to 2019–2020 appears in both products (correlation of the two ratio series, August–October, r = 0.91). With only two normal TROPOMI years, a TROPOMI version of the H4 anomaly test is not yet meaningful.

Other reflectance indices. EVI and NIRv, computed from the same MOD13Q1 composites, also declined before NDVI: NIRv in all nine years (mean +22.8 days season-scaled, +9.9 days decline-scaled) and EVI in nine and eight years (+17.1 and +7.7 days). These leads are as large as SIF's (+20.9 and +4.3 days). Compared directly, GOSIF did not decline before NIRv: it was first in 4 of 9 years season-scaled (mean −1.8 days) and in none decline-scaled (mean −6.1 days). For GOSIF, the lead over NDVI is therefore shared by a reflectance index of canopy structure and absorbed light, and comes from NDVI's late decline. GOSIF is itself predicted from MODIS reflectance (Section 3.1), so this agreement with NIRv is partly expected. TROPOMI, which is not, declined before NIRv in all three of its years, by 5.4, 7.8 and 13.1 days season-scaled and 4.6, 1.2 and 12.5 days decline-scaled (2018–2020), and before EVI by 3.0–20.8 days; across the four TROPOMI variants it came first against NIRv in 23 of 24 year-and-scaling cases. Three years, one of them a drought year, cannot establish a fluorescence-specific lead, but they do not support attributing all of SIF's lead to NDVI either.

4.7 Method Dependence and District-Level Lags (H2, H3)

Across the 40 combinations of Section 3.12 (all season-range), the claim that SIF leads in all nine years holds in all 8 mean-threshold combinations, in 4 of 8 at the 0.9 threshold, and in none of the 8 cross-correlation combinations. Using unweighted SIF and window-paired NDVI changes individual years' mean lag by −2.5 to +2.6 days; cropland weighting changes lags by at most 0.2 days. These choices matter far less than the scaling.

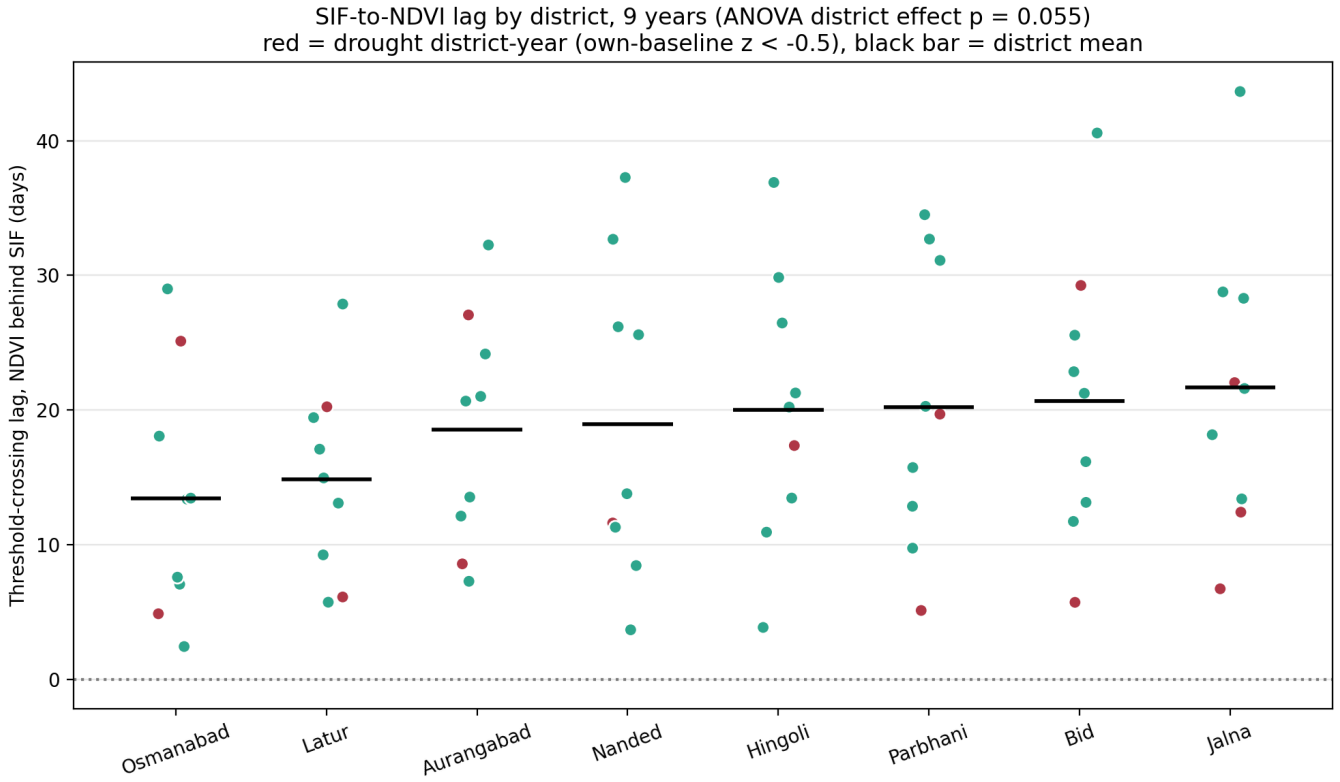


Figure 13. Season-range SIF–NDVI lag by district and year.

Under season-range scaling, SIF's decline came first in all 72 district-years (median 17.7 days, range 2.4–43.6), and the district effect is significant in the Friedman test (χ^2 = 14.5, p = 0.043) and borderline in the two-way ANOVA (p = 0.055). Under decline-range scaling, SIF is first in 52 of 72 district-years (median 2.3 days, range −8.1 to 19.9), district means range only from 2.9 to 5.6 days, and the Friedman test finds no district effect (p = 0.97). H2 is therefore not supported: the apparent district differences come from differences in how far NDVI stays up after the season.

H3 is not supported either. Regionally, drought years had a smaller season-range lag than normal years (14.8 vs 22.6 days) and a slightly larger decline-range lag (6.5 vs 3.6 days); with two drought years neither difference is interpretable. The block-bootstrap drought-minus-normal difference for the season-range lag is −7.8 days with a measurement-only interval of [−9.9, +1.8]. At district level (own baseline), drought district-years average 14.8 vs 19.5 days (season range, Welch p = 0.080) and 5.8 vs 3.4 days (decline range, p = 0.13).

4.8 Spatial SIF Pattern and Spatial Autocorrelation

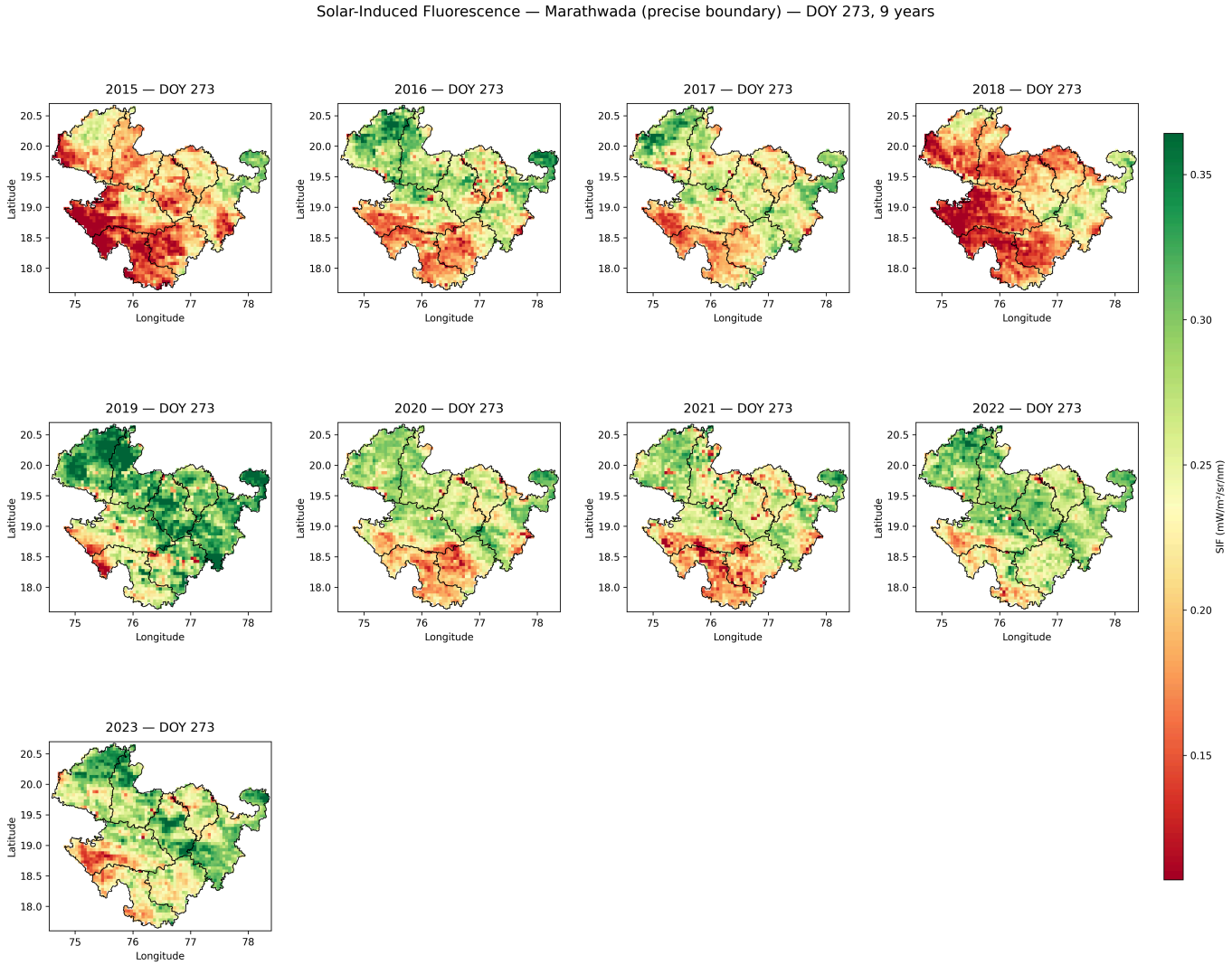


Figure 14. Spatial SIF distribution, day-of-year 273, 2015–2023.

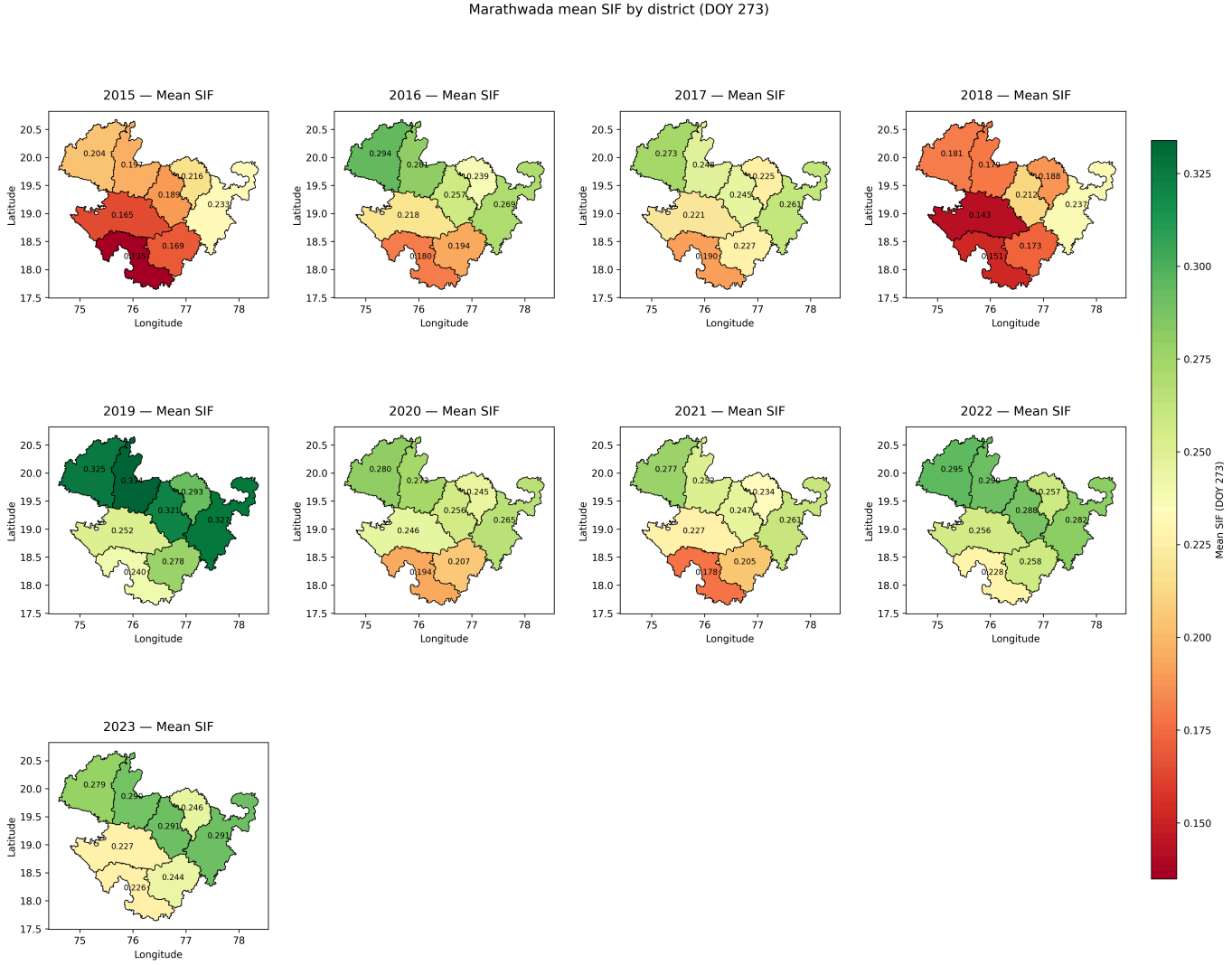


Figure 15. Mean SIF by district, day-of-year 273, 2015–2023.

Osmanabad has the lowest district SIF on day-of-year 273 in eight of nine years; in 2018 Bid (0.143) is lower than Osmanabad (0.151). The highest-SIF district varies: Aurangabad in five years, Nanded in two (2015, 2018), Jalna and Parbhani in one each.

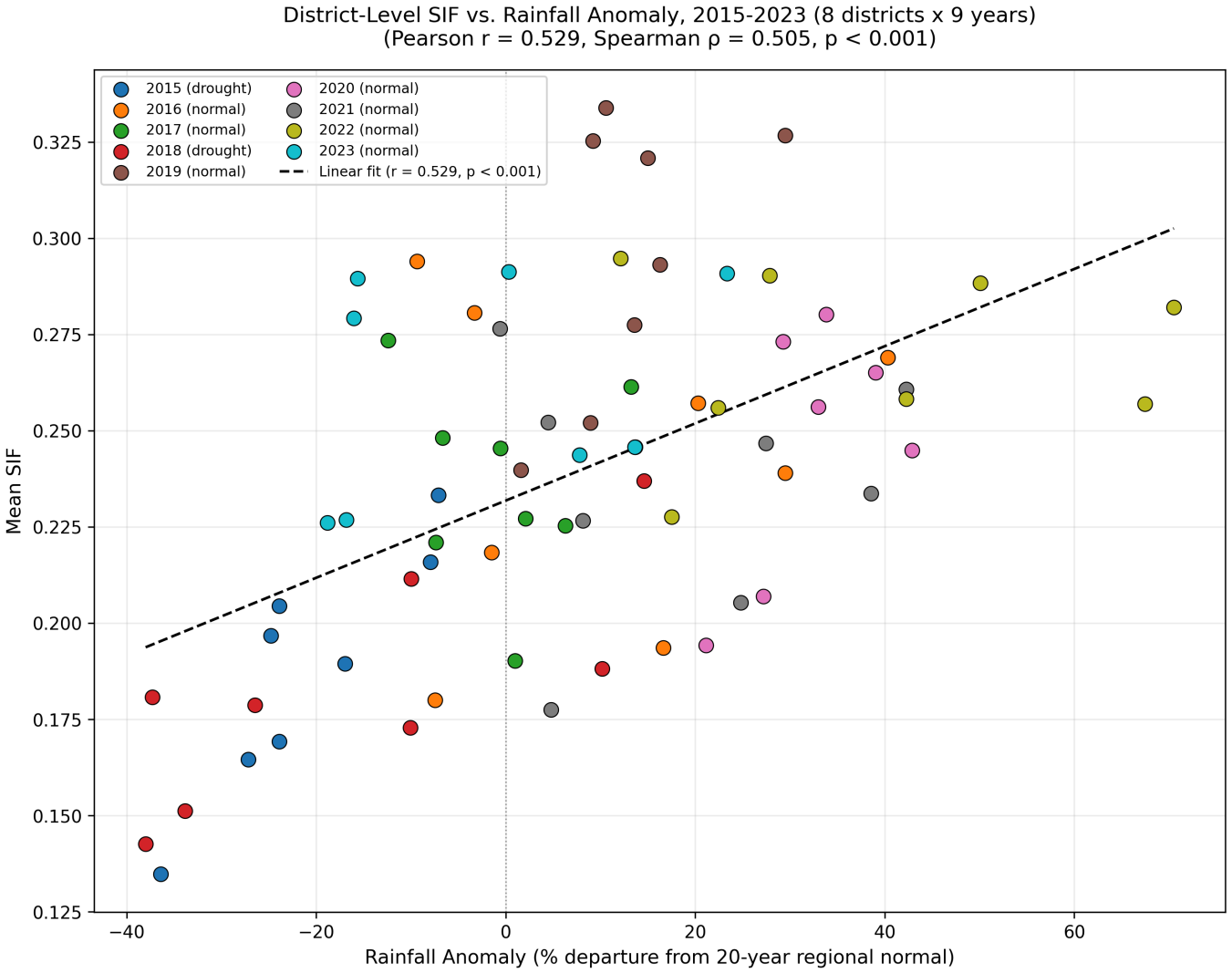


Figure 16. District-level SIF (day-of-year 273) against rainfall anomaly, 72 district-years.

District SIF and rainfall anomaly correlate across the 72 district-years (day-of-year 273: Pearson r = 0.53, Spearman ρ = 0.51, both p < 0.0001; p = 0.001 with standard errors clustered by year). Season-mean SIF correlates more strongly (r = 0.74), including within years (r = 0.66). With district and year fixed effects the relationship is not significant (p = 0.37 for season-mean SIF): SIF and rainfall move together across years and across districts, but a district's departure from its own average in a given year is not reliably tracked by SIF.

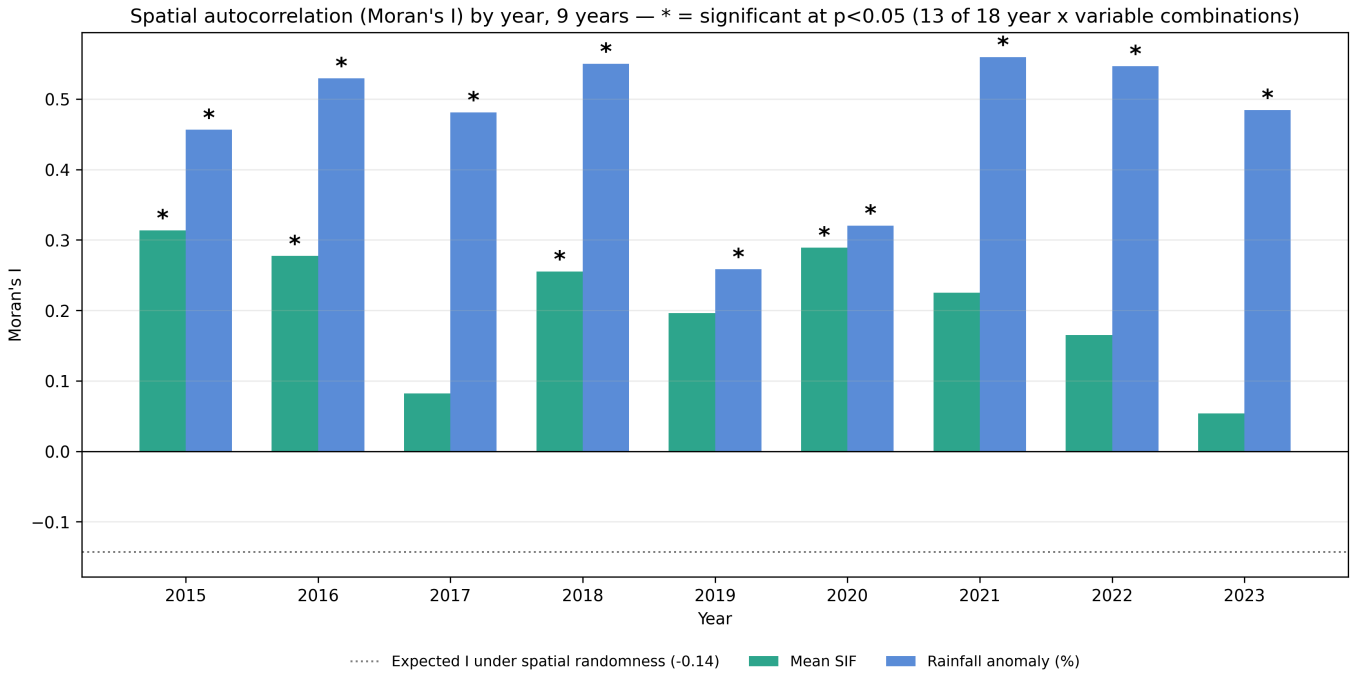


Figure 17. Moran's I for district SIF (day-of-year 273) and rainfall anomaly, by year (* = p < 0.05).

Rainfall anomaly is spatially clustered in all nine years (Moran's I 0.26–0.56). Day-of-year-273 SIF is clustered in four years, season-mean SIF in all nine (I = 0.29–0.54). District observations are therefore not independent, which is why the clustered and fixed-effects versions above are the ones to rely on.

4.9 Satellite Signals, the Rainfall Trigger and Official Declarations (RQ3)

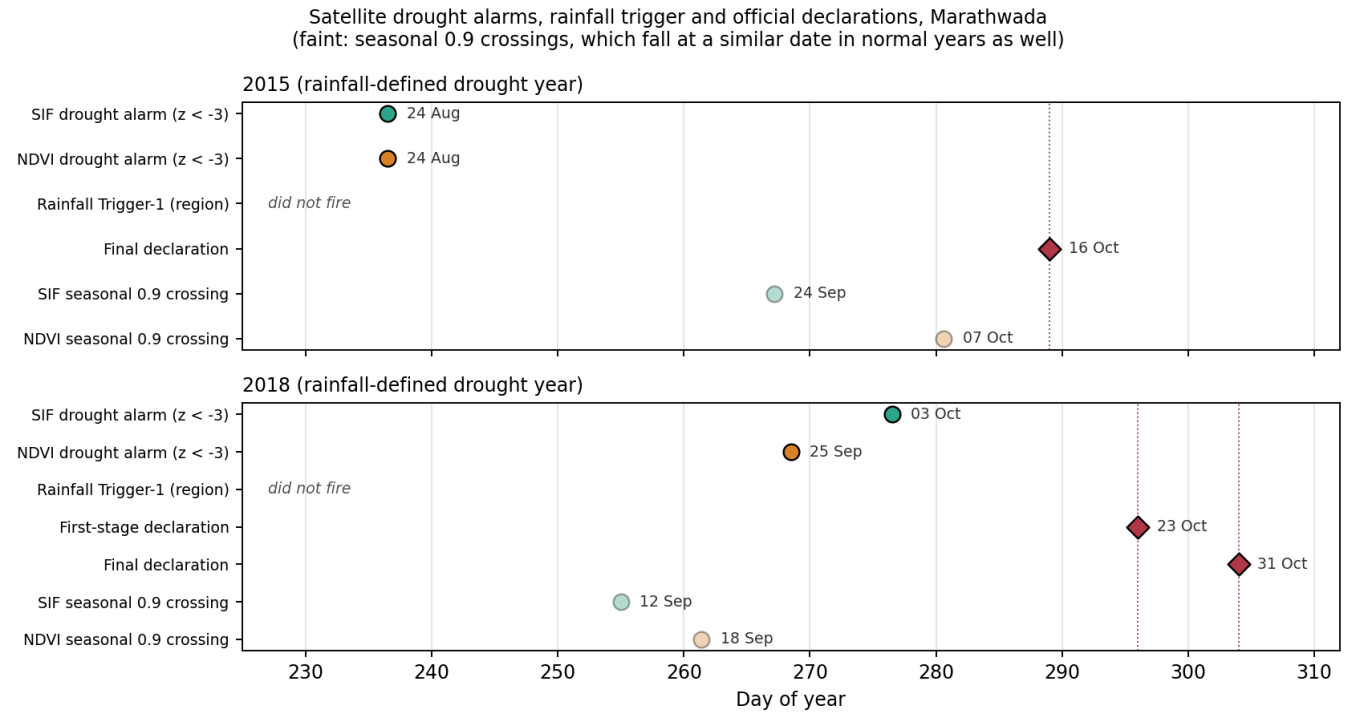


Figure 18. Timeline of the 2015 and 2018 seasons: SIF and NDVI drought alarms (z < −3), the regional Trigger-1 approximation (did not fire), the declaration dates, and, faintly, the seasonal 0.9 crossings.

Rainfall trigger. With dry spells counted inside the monsoon, the Trigger-1 approximation never fired for the region as a whole. At district level it fired in 2 of the 15 drought district-years — Aurangabad 2018 (3 October) and Bid 2018 (17 September) — and in one normal district-year (Bid 2017, 10 August). In Aurangabad both

alarms came 15.5 days before the trigger; in Bid 2018 the SIF alarm came 0.5 days and the NDVI alarm 8.5 days after it. The deficit criterion alone fires in 10 drought and 18 normal district-years and for the region in 2015, 2017, 2019 and 2023; the dry-spell criterion alone fires in 2 drought and 5 normal district-years. Counting dry spells to December, as an earlier version of this analysis did, makes the trigger fire in 4 drought district-years, mostly after the monsoon. Evaluated in every CHIRPS pixel, the result is the same: the trigger reached 25% of a district's area in only three district-years — Aurangabad 2018 (40%, 18 September), Bid 2018 (35%, 17 September) and Bid 2017, a normal year (72%, 10 August) — and in 2015, the stronger drought year, it fired in no more than 7% of any district. In Aurangabad 2018 both alarms came 0.5 days before the pixel-level trigger; in Bid 2018 the SIF alarm came 0.5 days and the NDVI alarm 8.5 days after it. The rarity of the trigger is therefore a property of the rule in this region, not of spatial averaging, and its deficit component alone separates drought from normal years only weakly: it fires in 10 of 15 drought but also in 18 of 57 normal district-years, and regionally in three normal years while missing 2018.

Declarations. The drought alarms ($z < -3$) came 52.5 days before the 2015 declaration (16 October) for both indices; 27.5 (SIF) and 35.5 (NDVI) days before the 2018 final declaration (31 October); and 19.5 and 27.5 days before the 2018 first-stage declaration (23 October). The seasonal 0.9 crossings (Figure 18, faint) happen at a similar time in every year, so they mark the normal start of the decline, not drought.

Under the Manual for Drought Management, the declaration follows a sequence of triggers that ends in field verification of crop damage (Ministry of Agriculture & Farmers Welfare, 2016a), so a satellite signal will precede it almost by construction. The more meaningful comparison is with the rainfall trigger. In the two cases where it fired, the satellite alarms came on the same date (Aurangabad) or SIF's one composite before NDVI's (Bid), and both were about as early as the trigger or earlier.

4.10 Crop-Yield Check

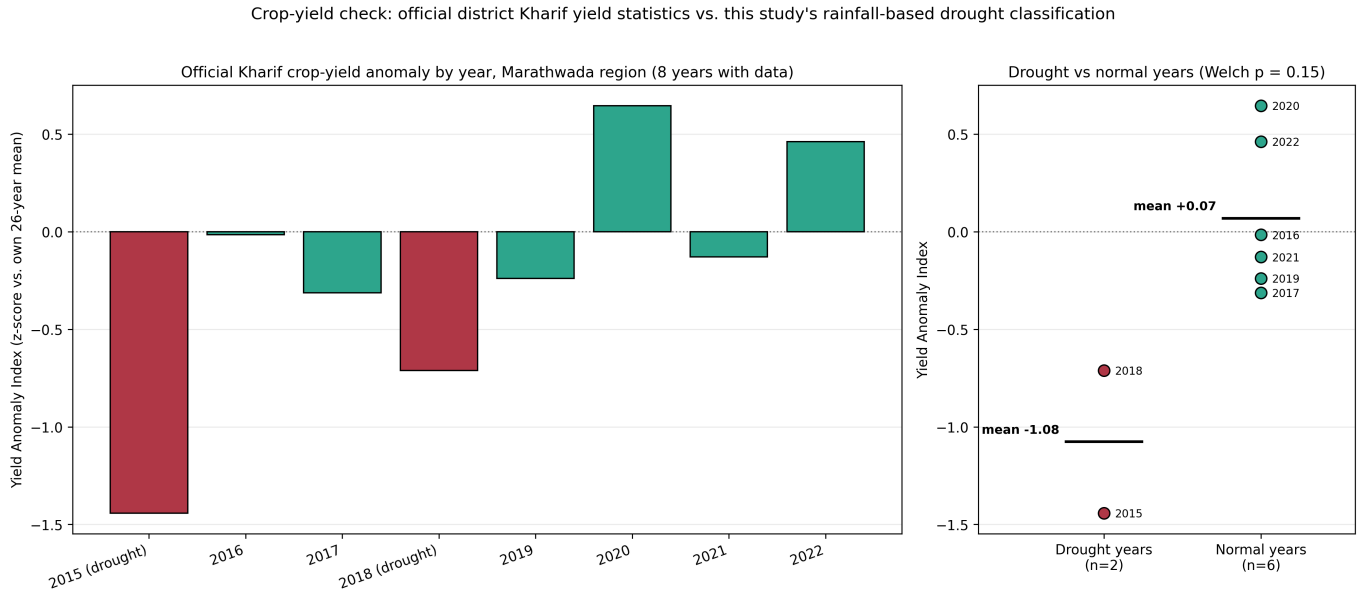


Figure 19. Marathwada Kharif Yield Anomaly Index by year (points per year and group means), with rainfall anomaly.

The Yield Anomaly Index (127 district-crop pairs, 16 crops) tracks rainfall closely ($r = 0.93$, $p = 0.0009$, $n = 8$). Its correlation with season-mean (June–December) regional SIF is $r = 0.88$ ($p = 0.004$), and $r = 0.80$–0.85 ($p = 0.007$–0.017) under versions without zero yields and outliers, area-weighted, detrended and major-crops-only; with day-of-year-273 SIF alone it is $r = 0.67$ ($p = 0.071$). At district level, with district and year fixed effects, the yield index remains associated with season-mean SIF ($p = 0.015$–0.040 across versions). The two drought years

had a mean index of −1.08 against +0.07 for the six normal years with data (Welch p = 0.15): lower harvests in the expected direction, but not significant with two drought years. This supports the drought classification; it says nothing about within-season timing.

4.11 Out-of-Sample Test, 2001–2014

Five of the fourteen years are drought years under the same rule: 2001 (z = −0.52, just past the cut-off), 2002 (−1.27), 2003 (−1.01), 2004 (−1.35) and 2014 (−1.12). Against each district's own baseline, 40 of the 112 district-years are drought district-years. The vegetation anomalies in these years were weaker than in 2015 and 2018: at z < −3, which again raised no alarm in any of the nine normal years, SIF flagged two of the five drought years and NDVI one (Table 5).

Table 5. Out-of-sample drought alarms, 2001–2014, with all methods unchanged. Region: 5 drought and 9 normal years. Districts (own rainfall baseline): 40 drought and 72 normal district-years. "First" counts a year in which only one index raised the alarm as that index being first.

Threshold	Region: drought flagged, SIF / NDVI	Region: first SIF / same / NDVI	Region: false alarms, SIF / NDVI	Districts: drought flagged, SIF / NDVI	Districts: false alarms, SIF / NDVI	Districts, both flagged: SIF earlier / same / NDVI earlier
z < −1	5 / 5	3 / 1 / 1	5 / 6	38 / 38	40 / 42	15 / 9 / 13
z < −1.5	5 / 4	4 / 1 / 0	2 / 3	25 / 25	16 / 25	5 / 11 / 6
z < −2	2 / 3	2 / 0 / 1	2 / 1	14 / 21	7 / 12	0 / 7 / 6
z < −3	2 / 1	1 / 1 / 0	0 / 0	3 / 6	0 / 1	0 / 1 / 1

At the regional scale the out-of-sample years lean towards SIF. At z < −3, SIF flagged 2001 (17 September) and 2002 (25 September), and NDVI only 2002, on the same date. At z < −1.5 SIF was first in four drought years and NDVI in none, with one tie: by 8 days in 2001 and 2002 and 64 days in 2014, and in 2004 NDVI did not reach the threshold. SIF also raised fewer false alarms at that threshold (2 against 3 of the 9 normal years).

At district level, where the sample is eight times larger, the picture matches 2015–2023: neither index is consistently first. NDVI flagged as many or more drought district-years at every threshold. Where both indices flagged one, SIF was earlier in 15 of 37 at z < −1, NDVI in 13, and 9 coincided; at z < −1.5 the counts were 5, 6 and 11, and at z < −2 0, 6 and 7. At z < −1.5, 17 of the 22 differences were at most one composite. SIF again raised fewer false alarms (16 against 25 at z < −1.5; 7 against 12 at z < −2). At z < −3 only two drought district-years were flagged by both: one on the same date, one with NDVI earlier. The regional series averages districts that differ in timing and in their normal-year spread, so regional and district alarms need not agree.

Over both periods together — seven drought years regionally and 55 drought district-years — SIF flagged four drought years at z < −3 and NDVI three; where both flagged, they coincided twice and NDVI was earlier once. Of the drought district-years flagged by both, SIF was earlier in 7 of 37 at z < −1.5, NDVI in 15, and 15 coincided; at z < −3, SIF was earlier in 7 of 14, NDVI in 4, and 3 coincided. H4 therefore remains unsupported: neither index is consistently earlier and most differences are one composite, although in 2001–2014 the regional series gave SIF a small edge.

Table 6. Regional seasonal lags in both periods (days; > 0 means the first-named series declines first). "First" is the number of years in which the first-named series declined first.

Comparison	Scaling	2015–2023: first	2015–2023: mean	2001–2014: first	2001–2014: mean	2001–2014: drought / normal mean
SIF vs NDVI	Season range	9 of 9	+20.9	14 of 14	+27.4	+28.3 / +26.9
SIF vs NDVI	Decline range	6 of 9	+4.3	13 of 14	+8.3	+7.6 / +8.6
NIRv vs NDVI	Season range	9 of 9	+22.8	14 of 14	+26.6	+28.9 / +25.3
NIRv vs NDVI	Decline range	9 of 9	+9.9	14 of 14	+13.8	+17.8 / +11.5
SIF vs NIRv	Season range	4 of 9	−1.8	10 of 14	+1.9	+0.2 / +2.9
SIF vs NIRv	Decline range	0 of 9	−6.1	3 of 14	−6.4	−12.4 / −3.0

The seasonal results reproduce (Table 6). Season-scaled, SIF declined before NDVI in all 14 years (mean +27.4 days), and NIRv did so by almost exactly as much (+26.6 days). Decline-scaled, SIF was first in 13 years (+8.3 days), but NIRv led NDVI by more (+13.8 days) and declined before SIF in 11 of the 14 years (mean −6.4 days). The lead over NDVI is again shared by a reflectance index, and it is not larger in drought years (season-scaled +28.3 against +26.9 days; decline-scaled +7.6 against +8.6), so H3 is not supported with five drought years either.

5. Discussion

The main result is negative for the question this study set out to answer. When each index is compared with its own normal-year behaviour, GOSIF SIF does not give a consistently earlier drought alarm than NDVI in Marathwada: at the regional scale the alarms coincide in 2015 and NDVI is one composite earlier in 2018, and across drought district-years the two are usually within one composite of each other, SIF more often first at the strictest threshold and NDVI at looser ones. A difference of one 8-day composite is below NDVI's 16-day sampling, and its direction reverses when NDVI is taken as observed rather than interpolated, so these data cannot resolve which index moves first. H4 is not supported.

The out-of-sample test on 2001–2014, run with every method frozen, mostly repeats this picture. At district level, with 40 more drought district-years, the alarms again mostly coincided or differed by one composite in either direction, and the seasonal lead again matched NIRv's and did not grow in drought. The regional alarms leaned towards SIF: it flagged one drought year (2001) in which NDVI did not reach the strict threshold, and it was first in four of five at a looser one. The regional series alone therefore cannot settle the question, but across 2001–2023 the evidence does not show a consistent SIF advantage. GOSIF is trained on OCO-2 soundings, which begin in September 2014, so for 2001–2014 it is also a prediction outside its training period.

The secondary result — SIF's seasonal decline starting before NDVI's — turns out to depend on how a decline is measured. Scaled over the whole season, SIF crosses every threshold first, in every year and every district-year. But NDVI in this region does not return to its pre-monsoon level after the season, while SIF largely does. On the season scale NDVI's decline is therefore measured against a floor it never reaches, and it crosses each threshold

later. Scaled over each series' own decline, the lead shrinks to 0–13 days and disappears in three years; the district differences disappear with it; and cross-correlation, which does not depend on scaling, agrees with the smaller estimate. The comparison with other reflectance indices shows where most of the remaining GOSIF lead comes from: NIRv and EVI, from the same MODIS composites, lead NDVI by as much as GOSIF does, and GOSIF does not lead NIRv. For GOSIF the lead is a property of NDVI — its saturation over dense canopy and its slow return to the pre-season level. TROPOMI, which does not use MODIS reflectance, declined before NIRv in all three of its years, by 1–13 days, so a small fluorescence-specific lead remains possible; with three years, one of them a drought year, it cannot be established.

Both results are consistent with a reading in which, at 5 km and 8- to 16-day resolution, GOSIF behaves largely as a greenness and absorbed-light signal. GOSIF is predicted from MODIS EVI and reanalysis weather (Li & Xiao, 2019), and canopy SIF is dominated by APAR and structure (Dechant et al., 2020), so it would be surprising if it carried an independent early stress signal; the NIRv result shows that, at these scales, it does not. Whether direct SIF retrievals do is open: the TROPOMI lead over NIRv is small and rests on three years, and more TROPOMI years are the obvious next test.

For policy, the relevant comparison is with the rainfall trigger, not the declaration. The trigger rarely fires, on averaged rainfall and pixel by pixel alike, and its deficit component alone fires in more normal than drought district-years (18 against 10) and missed the 2018 drought regionally. Satellite vegetation anomalies were available about as early as the trigger where it fired. That is an argument for vegetation anomalies as an input to drought monitoring; this study gives no reason to prefer GOSIF SIF over the NDVI already in use.

6. Limitations

- **Few drought years.** Only 2015 and 2018 meet the regional drought criterion in 2015–2023. The 2001–2014 test adds five more, but their vegetation anomalies were weaker, so at the strict alarm threshold it adds only two flagged drought years. Regional drought-versus-normal comparisons still rest on few seasons, and district-years within one season are not independent.
- **Normalisation dependence.** The seasonal lead depends on how each series is scaled (Section 4.4). The double-logistic fits that could provide a third, model-based estimate are unstable with the available sampling.
- **Lag size and sampling.** GOSIF is 8-day and MOD13Q1 16-day; decline-range leads are mostly smaller than the NDVI sampling interval.
- **Alarm timing resolution.** The SIF and NDVI drought alarms mostly differ by one 8-day composite or not at all, and the direction of that difference depends on whether NDVI is interpolated between observations or taken as observed (Section 4.2).
- **NDVI coverage.** Many monsoon NDVI composites rest on few clear pixels. The lag direction does not change when composites under 25% coverage are excluded, but with a 50% cut-off it reverses in 2022 (Supplementary Figure S1), and the timing of individual composites is uncertain.
- **Post-hoc thresholds.** The decline thresholds, the drought cut-off ($z < -0.5$) and the alarm threshold ($z < -3$) were chosen in the course of this study. Sensitivity checks are reported, and the 2001–2014 test applies them unchanged as a guard against tuning, but the analysis is exploratory.
- **Retrospective measures.** Both lag measures need the whole season, and the anomaly alarms use a normal-year reference that includes later years; they are retrospective timing comparisons, not an operational early-warning procedure.
- **GOSIF–MODIS dependence.** GOSIF uses MODIS reflectance as a predictor, which also makes its agreement with NIRv partly expected. TROPOMI removes this only for 2018–2020, with one drought year and two normal years. For 2001–2014, GOSIF is predicted outside the OCO-2 period it was trained on.

- **Spatial scale.** All results are regional or district averages at about 5 km over a mixed cropland mask; they do not describe individual crops or fields.
- **Rainfall trigger approximation.** Trigger-1 is applied by the state per taluka with field inputs; here it is approximated on district-averaged rainfall and per CHIRPS pixel, which give the same picture.
- **Declarations.** Only three declaration dates were found; the 2015 date is the date it was reported. The declaration process ends after field verification, so satellite signals precede it almost by construction.
- **Yield data.** Yield is one end-of-season number per district-crop, ends in 2022-23, and includes some crops grown partly outside the Kharif season.

7. Conclusion

Across nine Kharif seasons in Marathwada, GOSIF SIF did not flag agricultural drought resolvably earlier than NDVI. Compared with their own normal-year behaviour, the two indices raised the alarm on the same date in 2015 and NDVI one composite earlier in 2018, and across drought district-years they were usually within one composite of each other, with the direction depending on how NDVI is placed in time. Applied unchanged to 2001–2014, the same tests gave the same district-level picture and the same NIRv-shared seasonal lead; only the regional alarms leaned slightly towards SIF.

SIF's normal seasonal decline appears to start well before NDVI's when each curve is scaled over the whole season, but most of that lead comes from NDVI not returning to its pre-season level after the monsoon. Measured over each series' own decline, or by cross-correlation, the lead is small, varies between years, and is absent in about a third of them; and MODIS NIRv and EVI lead NDVI by as much as GOSIF does, so for GOSIF what remains is a property of NDVI, not of fluorescence. Direct TROPOMI SIF led NIRv slightly in its three years, which leaves a small fluorescence-specific lead open.

In this region and dataset, SIF is not an earlier drought alarm than NDVI at the resolution the data allow. Satellite vegetation anomalies may add lead time to a rainfall-led trigger, which rarely fired at these scales, but this study gives no evidence that GOSIF SIF would add more than NDVI already does. Testing that properly needs direct SIF retrievals over more drought seasons and vegetation data sampled more often than every 16 days.

Declarations

Ethics statement: Not applicable. This study did not involve human participants, animals, or personal data of any kind.

Competing interests: I declare no competing interests.

Funding: This research received no external funding. I covered all costs myself.

Data availability: Processed data and analysis code for this study are available at https://github.com/sakshimaske303-commits/GREEN_ALIBI; the full pipeline runs from `run_all.py` once the GOSIF files have been downloaded with the included download script. An interactive dashboard presenting this study's results is available at https://greenalibi-bzs2wvod5fflqh7dfe2cf4.streamlit.app. The raw satellite and rainfall datasets (GOSIF, TROPOSIF, MODIS, CHIRPS, FAO GAUL) are publicly available from their original providers, cited below.

CRediT author statement: Sakshi D. Maske: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization.

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Revision Note

This version corrects five data-handling errors present in the first preprint: NDVI dates one day early combined with nearest-date matching (which shortened every lag by about eight days); a cross-correlation search that could not return negative lags; a half-pixel longitude offset in the 2021 and 2022 GOSIF files; a district-name mismatch ("Beed" vs "Bid"); and two different rainfall sources used for different anomalies. Relative to the previous version it also: makes the drought-anomaly test (H4) the main question; adds the decline-range normalisation and double-logistic checks (Section 4.4), which change the interpretation of the seasonal lead and of H2; classifies district drought against each district's own rainfall baseline; counts dry spells for Trigger-1 only within the monsoon; uses a moving-block bootstrap centred on the reported lag; adds the NDVI coverage check; compares SIF with MODIS EVI and NIRv; evaluates Trigger-1 per CHIRPS pixel; and applies all methods unchanged to 2001–2014 as an out-of-sample test. A final check of this version found and corrected an alignment error in the district alarm comparison that dated every district NDVI alarm one composite (8 days) early; it also added the NDVI-dating sensitivity check (Section 3.9) and the TROPOMI–NIRv comparison (Section 4.6). GA_Development_Log.md records each change.

Supplementary Figures

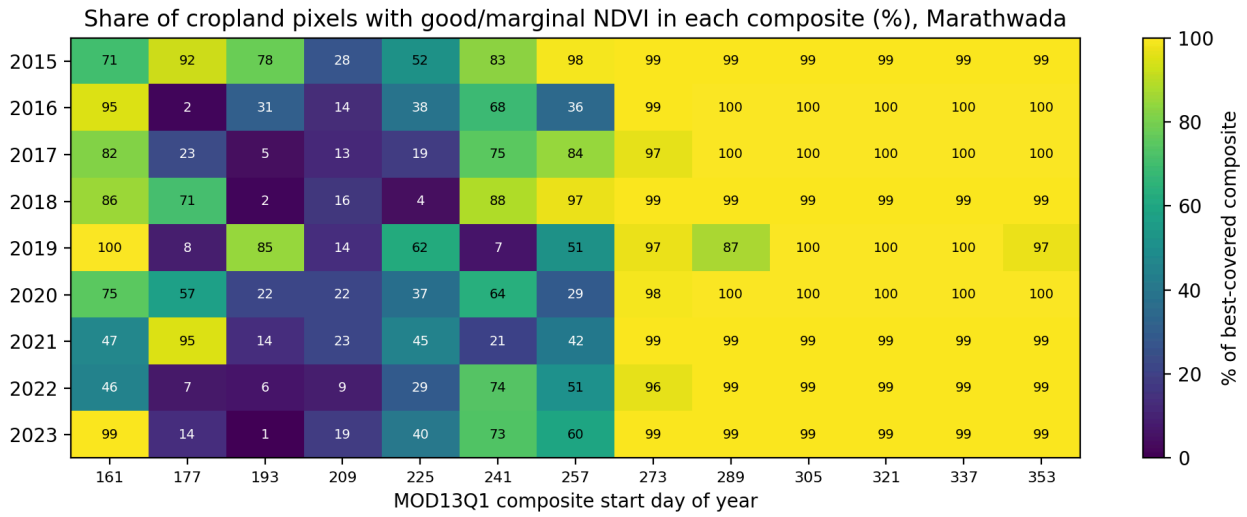


Figure S1. Share of cropland NDVI pixels passing the quality screen in each regional composite, and the regional season-range lag when composites below 10%, 25% or 50% coverage are excluded.

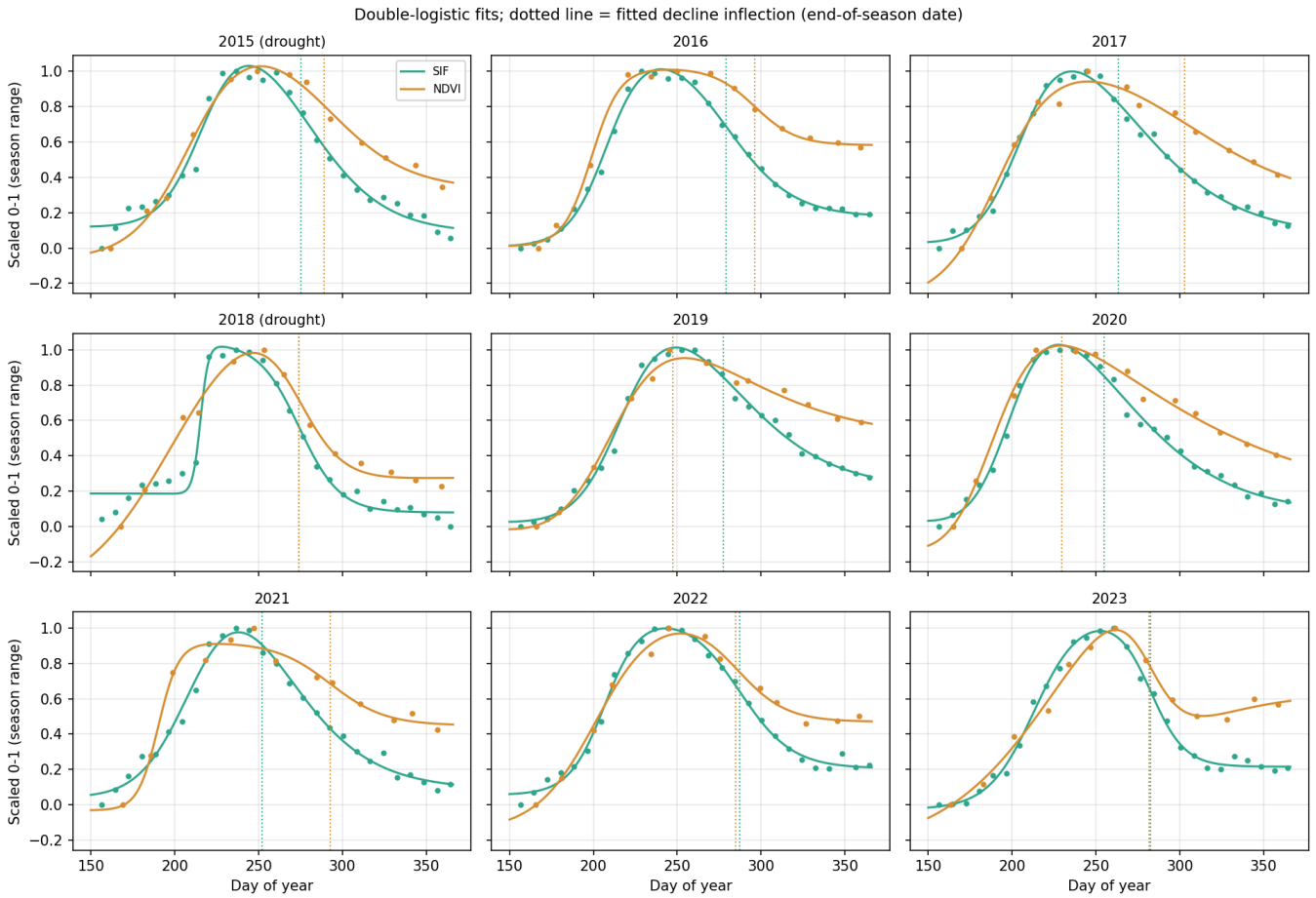


Figure S2. Asymmetric double-logistic fits to the regional SIF and NDVI series, by year, with the fitted end-of-season dates.

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